

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-12

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Action Center**Run date:** 2026-05-12 **Universe:** 668 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+20, 4.0%) **Clusters long:** 17/26 **ETFs long:** 502/668 (75%)**Today's events (cluster level)****BUY (1)**

cluster	theme	spread	p_h5	nss
cluster_5	BND	0.035	0.7	-0.177

Imminent (cluster level)Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; the p_h5 column is the 5-day probability for context.**Imminent BUY (top 5 by h=1 NSS)**

cluster	theme	spread	p_h1	p_h5	nss_h1	nss_h5
cluster_23	SHV + MINT + GBIL + ICSH	0.393	1	1	1	1.00
cluster_22	FLOT + BKLN + SRLN + FLRN + FLTR	1.305	1	1	1	1.00
cluster_3	QQQ	17.497	1	1	1	1.00
cluster_8	IEMG	11.512	1	1	1	1.00
cluster_4	VOO	8.861	1	1	1	0.99

Imminent SELL (top 3 by h=1 NSS)

cluster	theme	spread	p_h1	p_h5	nss_h1	nss_h5
cluster_26	SQQQ + UUP	-1.324	1.000	1.00	-1.000	-1.00
cluster_15	GLD + IAU + SLV	0.401	0.489	0.61	-0.343	-0.19
cluster_17	XLU + XLP	-0.034	0.649	0.74	-0.289	-0.23

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.95 fresh

Current state: LONG since 2026-04-13 (20 days, 4.0%)

5dma: 140.590 20dma: 138.541 50dma: 134.180

Spread vs 50dma: +6.410 Spread vs 20dma: +2.050

P(buy crossover within 5 days): 100.0% P(sell): 4.3%

Top 10 PC1 weights

	ticker	weight	name	category
102	ACWI	0.0021	iShares MSCI ACWI ETF	Global Equity
113	AOA	0.0021	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation
133	SPGM	0.0021	State Street SPDR Portfolio MSCI Global...	Global Equity
137	VT	0.0021	Vanguard Total World Stock Index ETF	Global Equity

46	SPY	0.0020	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity
116	AOR	0.0020	iShares Core 60/40 Balanced Allocation ...	Asset Allocation
120	URTH	0.0020	iShares MSCI World ETF	Global Equity
122	CRBN	0.0020	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity
180	RSP	0.0020	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity
264	IMCG	0.0020	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth

Position Sheet

Replicate each currently-long cluster via its 3-5 member basket. OLS weights sum to ~1 per cluster for unit-variance replication; scale each cluster's basket to your target cluster allocation. sticky=yes means members are unchanged from the prior run (only weights drift); sticky=no means a fresh greedy selection.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	weight	cor
QQQ	cluster_3	QQQ	1.000	20	15.3%	1.025	0.97
IEMG	cluster_8	IEMG	1.000	21	9.9%	0.931	0.99
SRLN	cluster_22	FLOT + BKLN + SRLN + FLRN +...	1.000	24	1.2%	0.311	0.89
FLOT	cluster_22	FLOT + BKLN + SRLN + FLRN +...	1.000	24	1.2%	0.292	0.89
BKLN	cluster_22	FLOT + BKLN + SRLN + FLRN +...	1.000	24	1.2%	0.209	0.89
FLRN	cluster_22	FLOT + BKLN + SRLN + FLRN +...	1.000	24	1.2%	0.127	0.89
FLTR	cluster_22	FLOT + BKLN + SRLN + FLRN +...	1.000	24	1.2%	0.034	0.89
VOO	cluster_4	VOO	0.987	20	5.9%	0.958	0.99
VTV	cluster_6	VTV	0.919	20	2.6%	1.021	0.98
PDBC	cluster_21	PDBC	0.918	84	47.8%	1.173	0.98
BWX	cluster_24	BWX	0.873	7	0.3%	0.752	0.93
KWEB	cluster_13	KWEB	0.858	18	2.8%	0.563	0.92
IJH	cluster_2	IJH	0.849	20	3.7%	1.009	0.99
VNQ	cluster_14	VNQ	0.816	21	4.2%	0.971	0.99
VTIP	cluster_19	VTIP	0.787	22	0.8%	1.724	0.90
SCHD	cluster_10	SCHD	0.787	14	0.6%	0.926	0.94
HYG	cluster_9	HYG	0.489	20	0.7%	0.911	0.95
IJR	cluster_7	IJR	0.380	21	2.9%	0.904	0.98
VHT	cluster_11	VHT	0.021	3	-2.0%	1.104	0.93
BND	cluster_5	BND	-0.177	0	0.0%	1.007	0.99
XLP	cluster_17	XLU + XLP	-0.232	6	-2.3%	0.588	0.99
XLU	cluster_17	XLU + XLP	-0.232	6	-2.3%	0.364	0.99

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
VEA	cluster_1	VEA	1.00	0.622	0.923	0.99
EWJ	cluster_20	EWJ	1.00	0.947	0.908	0.96
MUB	cluster_16	MUB	1.00	0.244	0.905	0.95
GUNR	cluster_12	URA + GUNR	1.00	0.550	0.897	0.95

ICSH	cluster_23	SHV + MINT + GBIL + ICSH	1.00	1.000	0.418	0.84
SHV	cluster_23	SHV + MINT + GBIL + ICSH	1.00	1.000	0.350	0.84
MINT	cluster_23	SHV + MINT + GBIL + ICSH	1.00	1.000	0.175	0.84
GBIL	cluster_23	SHV + MINT + GBIL + ICSH	1.00	1.000	0.117	0.84
URA	cluster_12	URA + GUNR	1.00	0.550	0.095	0.95
ILF	cluster_18	EWZ + ILF + DBA + ECH	0.98	0.072	0.382	0.93
EPI	cluster_25	INDA + INDY + EPI + PIN	0.98	0.084	0.256	1.00
INDA	cluster_25	INDA + INDY + EPI + PIN	0.98	0.084	0.250	1.00
PIN	cluster_25	INDA + INDY + EPI + PIN	0.98	0.084	0.249	1.00
INDY	cluster_25	INDA + INDY + EPI + PIN	0.98	0.084	0.246	1.00
ECH	cluster_18	EWZ + ILF + DBA + ECH	0.98	0.072	0.158	0.93
DBA	cluster_18	EWZ + ILF + DBA + ECH	0.98	0.072	0.056	0.93
EWZ	cluster_18	EWZ + ILF + DBA + ECH	0.98	0.072	0.047	0.93

Cluster Overview

name is the replication basket (post-NNLS, long-only). change is the signal-state shift vs the prior run; dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
3	cluster_3	QQQ	Technology	63	LONG	shifted	20	15.3%	1.000	+0.585
8	cluster_8	IEMG	Emerging Markets ...	25	LONG	shifted	21	9.9%	1.000	+0.595
22	cluster_22	FLOT + BKLN + SRLN + FLRN +...	Mixed	9	LONG	shifted	24	1.2%	1.000	+0.186
23	cluster_23	SHV + MINT + GBIL + ICSH	Cash / Ultra-Shor...	9	FLAT	shifted			1.000	+0.000
4	cluster_4	VOO	Mixed	61	LONG	shifted	20	5.9%	0.987	+1.183
20	cluster_20	EWJ	Japan Equity	9	FLAT	shifted			0.947	-0.014
6	cluster_6	VTV	Mixed	54	LONG	shifted	20	2.6%	0.919	-0.070
21	cluster_21	PDBC	Broad Commodities	9	LONG	shifted	84	47.8%	0.918	+0.102
24	cluster_24	BWX	Currency	6	LONG	shifted	7	0.3%	0.873	-0.127
13	cluster_13	KWEB	China Equity	15	LONG	shifted	18	2.8%	0.858	+1.200
2	cluster_2	IJH	Mixed	66	LONG	shifted	20	3.7%	0.849	+0.110
14	cluster_14	VNQ	Real Estate	14	LONG	shifted	21	4.2%	0.816	+0.852
10	cluster_10	SCHD	U.S. Dividend Equ...	20	LONG	shifted	14	0.6%	0.787	+0.644
19	cluster_19	VTIP	Inflation-Protect...	10	LONG	shifted	22	0.8%	0.787	+0.249
1	cluster_1	VEA	Mixed	82	FLAT	shifted			0.622	-0.378
12	cluster_12	URA + GUNR	Materials	17	FLAT	shifted			0.550	+0.517
9	cluster_9	HYG	High Yield Bond	22	LONG	shifted	20	0.7%	0.489	+1.459
7	cluster_7	IJR	Mixed	48	LONG	shifted	21	2.9%	0.380	-0.619
16	cluster_16	MUB	Municipal Bond	13	FLAT	shifted			0.244	-0.051
25	cluster_25	INDA + INDY + EPI + PIN	India Equity	4	FLAT	shifted			0.084	-0.916
18	cluster_18	EWZ + ILF + DBA + ECH	Mixed	11	FLAT	shifted			0.072	-0.138
11	cluster_11	VHT	Healthcare	17	LONG	shifted	3	-2.0%	0.021	-0.188
5	cluster_5	BND	Mixed	58	LONG	shifted	0	0.0%	-0.177	-1.089
15	cluster_15	GLD + IAU + SLV	Mixed	13	FLAT	shifted			-0.193	-0.964
17	cluster_17	XLU + XLP	Consumer Staples	11	LONG	shifted	6	-2.3%	-0.232	-1.025
26	cluster_26	SQQQ + UUP	Currency	2	FLAT	shifted			-1.000	-1.791

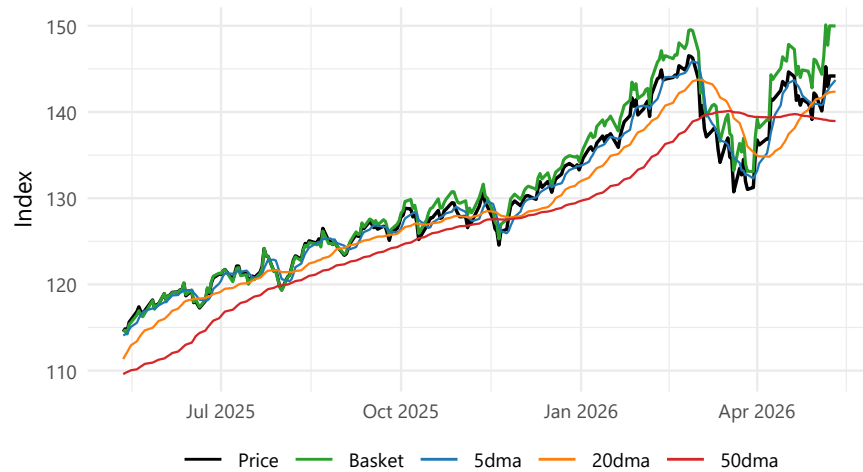
Cluster Details

Cluster 1 – VEA

Members: 82 **Var explained:** 0.839 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +44.8%)
- Latest sell_cross: 2026-05-04
- Current 252d return: +27.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
International Equity	21	0.2686
Developed ex-US Equity	9	0.1179
International Dividend Equity	6	0.0748
Global ex-US Equity	5	0.0656
Europe Equity	4	0.0518
Alternatives	4	0.0464
Eurozone Equity	3	0.0372
Asia-Pacific Equity	2	0.0242

Real Estate	2	0.0225
Global Equity	2	0.0222
Emerging Markets Fixed Income	2	0.0193
Currency	2	0.0181
Commodity	1	0.0132
Unknown	1	0.0130
Global ex-US Small Cap Equity	1	0.0129
Germany Equity	1	0.0124
France Equity	1	0.0124
United Kingdom Equity	1	0.0123
Italy Equity	1	0.0122
Europe Financials	1	0.0120
Sweden Equity	1	0.0120
Netherlands Equity	1	0.0120
Australia Equity	1	0.0119
Spain Equity	1	0.0116
Canada Equity	1	0.0114
Switzerland Equity	1	0.0114
Belgium Equity	1	0.0113
Austria Equity	1	0.0112
Poland Equity	1	0.0104
South Africa Equity	1	0.0099
New Zealand Equity	1	0.0091
Mexico Equity	1	0.0089

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.003

ticker	weight_ols	name	category	etf_signal
VEA	0.923	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members

	ticker	weight	name	category	etf_signal
2	EFA	0.0133	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0133	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
63	IEFA	0.0133	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
64	SCHF	0.0133	Schwab International Equity ETF	Developed ex-US Equity	LONG
65	SPDW	0.0133	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
21	DWM	0.0132	WisdomTree International Equity Fund	International Dividend E...	FLAT
34	TLTD	0.0132	FlexShares Morningstar Developed Market...	International Equity	FLAT
49	GSIE	0.0132	Goldman Sachs ActiveBeta International ...	Commodity	FLAT

52	ESGD	0.0132	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
28	PXF	0.0131	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
38	ACWX	0.0131	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
39	CWI	0.0131	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
41	INTF	0.0131	iShares International Equity Factor ETF	International Equity	FLAT
42	IQLT	0.0131	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
44	QEFA	0.0131	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
58	IEUR	0.0131	iShares Core MSCI Europe ETF	Europe Equity	FLAT
59	IXUS	0.0131	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
60	VEU	0.0131	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
61	VGK	0.0131	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
62	VXUS	0.0131	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
6	IEV	0.0130	iShares Europe ETF	Europe Equity	FLAT
32	EFV	0.0130	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
35	JPIN	0.0130	JPMorgan Diversified Return Internation...	International Equity	FLAT
46	RODM	0.0130	Hartford Multifactor Developed Markets ...	International Equity	FLAT
47	VYMI	0.0130	Vanguard International High Dividend Yi...	International Dividend E...	LONG
48	FNDF	0.0130	Schwab Fundamental International Equity...	International Equity	LONG
56	SCHC	0.0130	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
75	FEU	0.0130			FLAT
24	IQDF	0.0129	FlexShares International Quality Divide...	International Equity	LONG
31	SCZ	0.0129	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
33	FNDC	0.0129	Schwab Fundamental International Small ...	International Equity	LONG
57	VSS	0.0129	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
8	DLS	0.0128	WisdomTree International SmallCap Fund	International Equity	LONG
19	EZU	0.0128	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
43	IVLU	0.0128	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
72	HFXI	0.0128	NYLI FTSE International Equity Currency...	International Equity	LONG
29	EFG	0.0127	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
45	FEZ	0.0127	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
54	VIGI	0.0127	Vanguard International Dividend Appreci...	International Dividend E...	LONG
4	FEP	0.0126	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
30	GWX	0.0126	State Street SPDR S&P International Sma...	International Equity	LONG
40	IMTM	0.0126	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
1	EWG	0.0124	iShares MSCI Germany Index Fund	Germany Equity	LONG
5	VIDI	0.0124	Vident International Equity Strategy ETF	International Equity	LONG
16	EWQ	0.0124	iShares MSCI France Index Fund	France Equity	FLAT
20	IDV	0.0124	iShares International Select Dividend E...	International Dividend E...	LONG
53	HDEF	0.0124	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
17	EWU	0.0123	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
50	IDLV	0.0123	Invesco S&P International Developed Low...	International Equity	LONG
79	IPKW	0.0123	Invesco International BuyBack Achievers...	International Equity	LONG
82	EWI	0.0122	iShares MSCI Italy ETF	Italy Equity	LONG
22	EPP	0.0121	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
51	EFAV	0.0121	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT

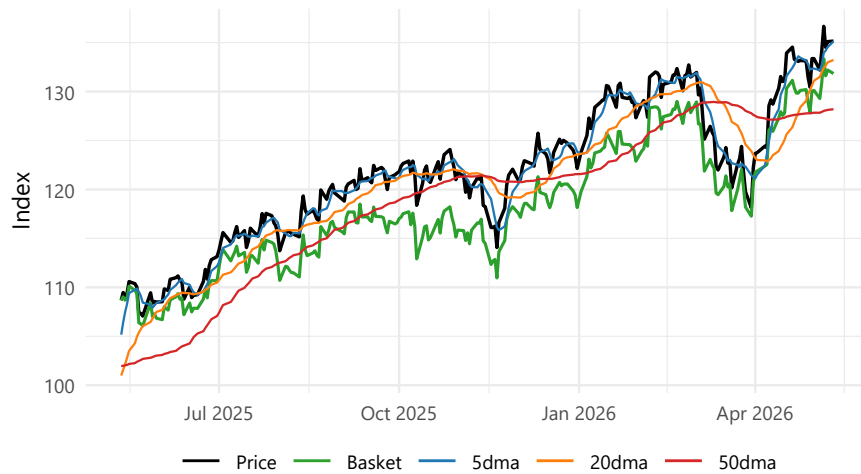
23	EUFN	0.0120	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
71	EWD	0.0120	iShares MSCI Sweden ETF	Sweden Equity	LONG
77	EWN	0.0120	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
81	BBAX	0.0120	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
13	EWA	0.0119	iShares MSCI Australia Index Fund	Australia Equity	FLAT
27	DWX	0.0119	State Street SPDR S&P International Div...	International Dividend E...	LONG
12	HEZU	0.0118	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
36	DBEF	0.0118	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
37	HEFA	0.0118	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
11	PID	0.0117	Invesco International Dividend Achiever...	International Dividend E...	LONG
15	EWP	0.0116	iShares MSCI Spain ETF	Spain Equity	FLAT
9	HEDJ	0.0114	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
10	IHDG	0.0114	WisdomTree International Hedged Quality...	Alternatives	LONG
14	EWL	0.0114	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
18	EWG	0.0114	iShares MSCI Canada Index Fund	Canada Equity	LONG
25	NFRA	0.0114	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
55	VNQI	0.0114	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
80	EWK	0.0113	iShares MSCI Belgium ETF	Belgium Equity	FLAT
76	EWO	0.0112	iShares MSCI Austria ETF	Austria Equity	LONG
7	RWX	0.0111	State Street SPDR Dow Jones Internation...	Real Estate	LONG
26	IGF	0.0108	iShares Global Infrastructure ETF	Global Equity	LONG
70	EPOL	0.0104	iShares MSCI Poland ETF	Poland Equity	FLAT
74	EZA	0.0099	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
69	FXA	0.0097	Invesco CurrencyShares Australian Dolla...	Currency	LONG
66	EBND	0.0096	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
67	EMLC	0.0096	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
73	ENZL	0.0091	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
68	EWX	0.0089	iShares MSCI Mexico ETF	Mexico Equity	LONG
78	FXC	0.0084	Invesco CurrencyShares Canadian Dollar ...	Currency	LONG

Cluster 2 – IJH

Members: 66 **Var explained:** 0.848 **Signal:** LONG since 2026-04-13 (20 days, 3.7%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +31.7%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +28.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Industrials	15	0.2179
U.S. Small Cap Growth	6	0.0954
U.S. Mid Cap Growth	6	0.0949
Consumer Discretionary	6	0.0845
U.S. Mid Cap Equity	5	0.0813
U.S. Small Cap Equity	5	0.0803
Utilities	3	0.0428
U.S. Momentum Equity	3	0.0428
Growth Equity	2	0.0307
U.S. Small Cap Momentum Equity	2	0.0304
Momentum Equity	2	0.0296
Unknown	1	0.0163
U.S. Small Cap ESG Equity	1	0.0162
U.S. Small Cap Multifactor Equity	1	0.0161
U.S. Extended Market Equity	1	0.0161
U.S. Mid Cap Quality Equity	1	0.0158
U.S. Mid Cap Momentum Equity	1	0.0154
U.S. Small Cap Industrials	1	0.0151
Technology	1	0.0150
U.S. Micro Cap Equity	1	0.0150
IPO Equity	1	0.0143

U.S. Small Cap Consumer Discretionary 1 0.0142

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.004

ticker	weight_ols	name	category	etf_signal
IJH	1.009	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
23	SCHM	0.0163	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
29	MDY	0.0163	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	LONG
37	IVOO	0.0163	Vanguard S&P Mid-Cap 400 ETF		LONG
42	IJH	0.0163	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	SPMD	0.0163	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	LONG
44	VB	0.0163	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG
20	JHMM	0.0162	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	LONG
26	NUSC	0.0162	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	LONG
31	IJK	0.0162	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
33	IVOG	0.0162	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
34	MDYG	0.0162	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
45	SCHA	0.0162	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
27	SMLF	0.0161	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
41	VXF	0.0161	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
4	IWM	0.0160	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
30	IJT	0.0160	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	LONG
32	VIOG	0.0160	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	LONG
35	SLYG	0.0160	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	LONG
38	VTWO	0.0160	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
40	ISCG	0.0159	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
15	FXR	0.0158	First Trust Industrials AlphaDEX	Industrials	LONG
28	IWO	0.0158	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
36	VTWG	0.0158	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
39	VBK	0.0158	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
54	PSC	0.0158	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
61	XMHQ	0.0158	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	LONG
14	IMCG	0.0157	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
21	VIS	0.0157	Vanguard Industrials ETF	Industrials	LONG
19	IYJ	0.0156	iShares U.S. Industrials ETF	Industrials	LONG
22	FIDU	0.0156	Fidelity MSCI Industrials Index ETF	Industrials	LONG
60	PAVE	0.0155	Global X U.S. Infrastructure Developmen...	Industrials	LONG

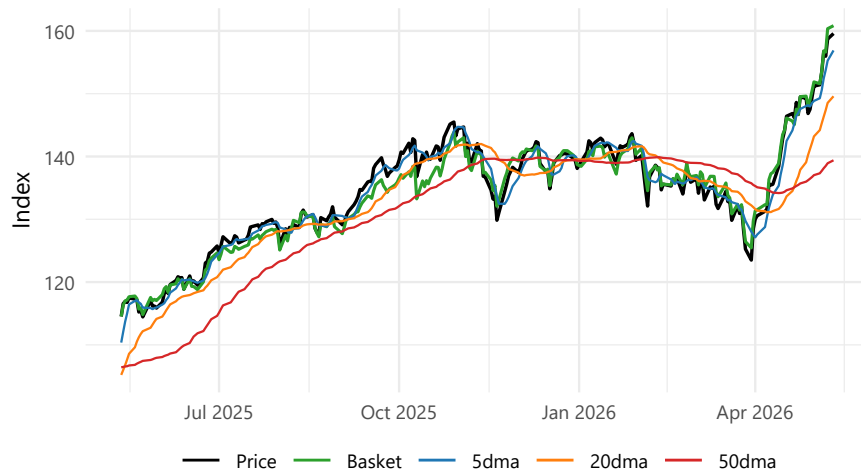
10	XMMO	0.0154	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
11	IWP	0.0153	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
13	VOT	0.0153	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
46	FXD	0.0153	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
62	VFMO	0.0153	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
66	XSMO	0.0153	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG
1	XLI	0.0152	State Street Industrial Select Sector S...	Industrials	LONG
59	PSCI	0.0151	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	LONG
6	FTC	0.0150	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
24	DWAS	0.0150	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
25	IWC	0.0150	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
53	RSPD	0.0150	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
7	PDP	0.0149	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
9	RPG	0.0148	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
18	FIW	0.0148	First Trust Water ETF	Utilities	LONG
48	FTXR	0.0148	First Trust Nasdaq Transportation ETF	Industrials	LONG
5	FV	0.0147	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
63	PKB	0.0147	Invesco Building & Construction ETF	Industrials	LONG
64	PRN	0.0146	Invesco Dorsey Wright Industrials Momen...	Industrials	LONG
16	PHO	0.0145	Invesco Water Resources ETF	Utilities	FLAT
8	FPX	0.0143	First Trust US Equity Opportunities ETF	IPO Equity	LONG
57	XTN	0.0143	State Street SPDR S&P Transportation ETF	Industrials	FLAT
47	PEJ	0.0142	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
65	PSCD	0.0142	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
52	IYT	0.0141	iShares U.S. Transportation ETF	Industrials	LONG
2	XRT	0.0139	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
58	QMOM	0.0138	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
12	MTUM	0.0137	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
56	XHB	0.0137	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
17	CGW	0.0135	Invesco S&P Global Water Index ETF	Utilities	FLAT
55	XAR	0.0135	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
49	PPA	0.0133	Invesco Aerospace & Defense ETF	Industrials	FLAT
3	JETS	0.0127	U.S. Global Jets ETF	Industrials	LONG
50	ITA	0.0125	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT
51	ITB	0.0124	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 3 – QQQ

Members: 63 **Var explained:** 0.840 **Signal:** LONG since 2026-04-13 (20 days, 15.3%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +34.2%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +44.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Technology	34	0.5348
U.S. Large Cap Growth	11	0.1854
U.S. Equity	2	0.0333
Growth Equity	2	0.0332
Unknown	2	0.0315
Convertible Bond	2	0.0306
Financials	2	0.0298
U.S. Mega Cap Growth	1	0.0167
U.S. Equal Weight Equity	1	0.0167
U.S. Small Cap Technology	1	0.0153
IPO Equity	1	0.0152
Options Income	1	0.0152
Trend-Following Equity	1	0.0148
Consumer Discretionary	1	0.0147
U.S. Trend-Following Equity	1	0.0128

Replication basket – QQQ (cor=0.97) basket lead at lag 1: -0.014

ticker	weight_ols	name	category	etf_signal
QQQ	1.025	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members

	ticker	weight	name	category	etf_signal
14	QQQ	0.0170	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
35	IGM	0.0170	iShares Expanded Tech Sector ETF	Technology	LONG
40	ONEQ	0.0170	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
50	ILCG	0.0170	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
29	XNTK	0.0169	State Street SPDR NYSE Technology ETF	Technology	LONG
39	NULG	0.0169	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
43	IWF	0.0169	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
45	VGT	0.0169	Vanguard Information Tech ETF	Technology	LONG
47	FTEC	0.0169	Fidelity MSCI Information Technology In...	Technology	LONG
48	VONG	0.0169	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
52	SCHG	0.0169	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
54	VUG	0.0169	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
8	AIQ	0.0168	Global X Artificial Intelligence & Tech...	Technology	LONG
18	QTEC	0.0168	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
24	FXL	0.0168	First Trust Technology AlphaDEX	Technology	LONG
37	IYW	0.0168	iShares U.S. Technology ETF	Technology	LONG
44	IVW	0.0168	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
51	IUSG	0.0168	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
17	QQEW	0.0167	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
36	IXN	0.0167	iShares Global Tech ETF	Technology	LONG
42	IWY	0.0167	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
46	VOOG	0.0167	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
49	MGK	0.0167	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
53	SPYG	0.0167	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0166	State Street Technology Select Sector S...	Technology	LONG
21	XT	0.0165	iShares Future Exponential Technologies...	Technology	LONG
34	PWB	0.0165	Invesco Large Cap Growth ETF	Growth Equity	LONG
55	SPHB	0.0162	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
20	TDIV	0.0160	First Trust NASDAQ Technology Dividend ...	Technology	LONG
28	FDN	0.0160	First Trust DJ Internet Index Fund	Technology	LONG
60	FIVG	0.0160			FLAT
6	BOTZ	0.0159	Global X Robotics & Artificial Intellig...	Technology	LONG
7	ROBO	0.0159	ROBO Global Robotics and Automation Ind...	Technology	LONG
63	QTUM	0.0159	Defiance Quantum ETF	Technology	LONG

27	PNQI	0.0158	Invesco Nasdaq Internet ETF	Technology	LONG
5	SKYY	0.0157	First Trust Cloud Computing ETF	Technology	LONG
2	SOXX	0.0156	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
3	SMH	0.0156	VanEck Semiconductor ETF	Technology	LONG
4	IGV	0.0155	iShares Expanded Tech-Software Sector E...	Technology	LONG
22	XSD	0.0155	State Street SPDR S&P Semiconductor ETF	Technology	LONG
38	CWB	0.0155	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
61	PBS	0.0155			LONG
62	PTF	0.0155	Invesco Dorsey Wright Technology Moment...	Technology	LONG
13	ARKQ	0.0154	ARK Autonomous Technology & Robotics ETF	Technology	LONG
15	NXTG	0.0154	First Trust Indxx NextG ETF	Technology	LONG
19	PSI	0.0153	Invesco Semiconductors ETF	Technology	LONG
57	PSCT	0.0153	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
26	IPO	0.0152	Renaissance IPO ETF	IPO Equity	LONG
31	FINX	0.0152	Global X FinTech ETF	Financials	LONG
33	QYLD	0.0152	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
59	FTXL	0.0152	First Trust Nasdaq Semiconductor ETF	Technology	LONG
12	ARKW	0.0151	ARK Next Generation Internet ETF	Technology	LONG
41	ICVT	0.0151	iShares Convertible Bond ETF	Convertible Bond	LONG
11	ARKK	0.0148	ARK Innovation ETF	Technology	LONG
25	CIBR	0.0148	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
32	PTNQ	0.0148	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
23	IBUY	0.0147	Amplify Online Retail ETF	Consumer Discretionary	FLAT
10	HACK	0.0146	Amplify Cybersecurity ETF	Technology	LONG
30	IPAY	0.0146	Amplify Digital Payments ETF	Financials	FLAT
16	PRNT	0.0144	3D Printing (The) ETF	Technology	LONG
9	BLOK	0.0137	Amplify Blockchain Technology ETF	Technology	LONG
56	PTLC	0.0128	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
58	BLCN	0.0123	Siren NexGen Economy ETF	Technology	LONG

Cluster 4 – VOO

Members: 61 **Var explained:** 0.892 **Signal:** LONG since 2026-04-13 (20 days, 5.9%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.5%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +28.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1558
U.S. Total Market Equity	7	0.1214
Global Equity	7	0.1191
Consumer Discretionary	6	0.0914
Asset Allocation	4	0.0631
Communication Services	4	0.0592
U.S. Mega Cap Equity	3	0.0510
U.S. Quality / Factor Equity	3	0.0505
U.S. Multifactor Equity	3	0.0504
U.S. ESG Equity	2	0.0345
U.S. Equity	2	0.0345
Unknown	2	0.0312
Commodity	1	0.0173
Large Cap Value Equities	1	0.0172
Large Cap Blend Equities	1	0.0171
Dividend Equity	1	0.0168
U.S. Dividend Equity	1	0.0165
Alternatives	1	0.0149
International IPO Equity	1	0.0133
Israel Equity	1	0.0126
Singapore Equity	1	0.0121

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.005

ticker	weight_ols	name	category	etf_signal
VOO	0.958	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
37	IWB	0.0174	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
49	ITOT	0.0174	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
51	SCHB	0.0174	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
4	SPY	0.0173	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
12	URTH	0.0173	iShares MSCI World ETF	Global Equity	LONG
30	IWV	0.0173	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
31	IYY	0.0173	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
34	SPYX	0.0173	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
36	ESGU	0.0173	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
42	VTHR	0.0173	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
43	GSLC	0.0173	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
45	VONE	0.0173	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
47	VV	0.0173	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
48	ILCB	0.0173	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
50	IVV	0.0173	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
52	SCHX	0.0173	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
53	SPLG	0.0173			LONG
54	SPTM	0.0173	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
55	VOO	0.0173	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
56	VTI	0.0173	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
26	JHML	0.0172	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
29	SUSA	0.0172	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
38	IWL	0.0172	iShares Russell Top 200 ETF	U.S. Equity	LONG
44	EPS	0.0172	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
46	MGC	0.0172	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
7	ACWI	0.0171	iShares MSCI ACWI ETF	Global Equity	LONG
13	CRBN	0.0171	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
15	VT	0.0171	Vanguard Total World Stock Index ETF	Global Equity	LONG
22	TILT	0.0171	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
28	DSI	0.0171	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
32	LRGF	0.0171	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
8	AOA	0.0170	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
14	SPGM	0.0170	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
33	OEF	0.0170	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG

39	QUAL	0.0170	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
24	QDF	0.0168	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
40	QUS	0.0168	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
23	I00	0.0167	iShares Global 100 ETF	Global Equity	LONG
25	CATH	0.0167	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	XLG	0.0167	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
41	SPHQ	0.0166	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0165	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
27	DGRW	0.0165	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
17	IYC	0.0162	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
60	OMFL	0.0161	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
21	FDIS	0.0159	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
18	VCR	0.0158	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
1	XLY	0.0156	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0155	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
19	VOX	0.0153	Vanguard Communication Services ETF	Communication Services	LONG
20	FCOM	0.0153	Fidelity MSCI Communication Services In...	Communication Services	LONG
2	XLC	0.0149	State Street Communication Services Sel...	Communication Services	LONG
5	QAI	0.0149	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
59	RTH	0.0147	VanEck Retail ETF	Consumer Discretionary	LONG
9	AOK	0.0142	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
61	IGN	0.0139			FLAT
57	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	FLAT
16	FPXI	0.0133	First Trust International Equity Opport...	International IPO Equity	LONG
3	ESPO	0.0132	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
58	EIS	0.0126	iShares MSCI Israel ETF	Israel Equity	LONG
6	EWS	0.0121	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 5 – BND

Members: 58 **Var explained:** 0.832 **Signal:** LONG since 2026-05-11 (0 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-05-11 – filter **PASS** (252d at date = +5.1%)
- Latest sell_cross: 2026-04-30
- Current 252d return: +5.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-11 > last sell at 2026-04-30)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3282
Investment Grade Corporate Bond	13	0.2301
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0890
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0309
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.007

ticker	weight_ols	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members

	ticker	weight	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	LONG
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
46	SCHR	0.0182	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	LONG
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	LONG
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
50	MBB	0.0180	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
52	SPMB	0.0179	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	LONG
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

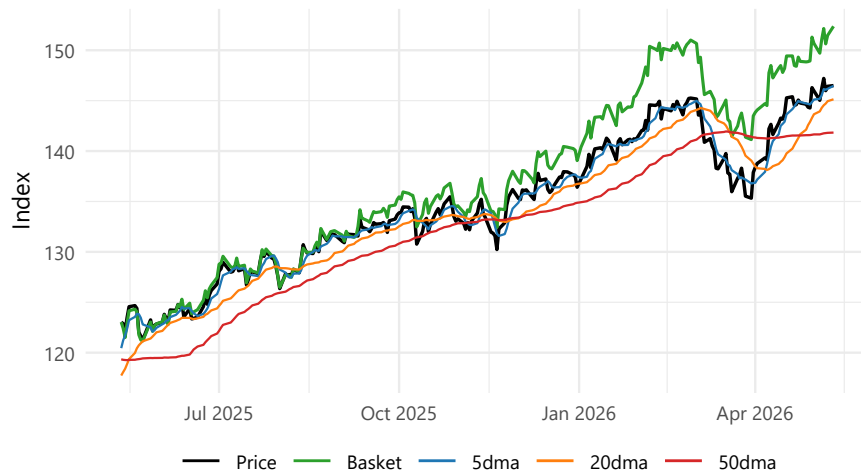
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	LONG
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0157	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0152	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0136	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0120	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 6 – VTV

Members: 54 **Var explained:** 0.922 **Signal:** LONG since 2026-04-13 (20 days, 2.6%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +23.0%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +21.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	11	0.2040
U.S. Large Cap Value	10	0.1903
U.S. Multifactor Equity	3	0.0559
U.S. Mid Cap Equity	3	0.0556
U.S. Value Factor Equity	3	0.0553
U.S. Low Volatility Equity	3	0.0514
U.S. Fundamental Equity	2	0.0380
U.S. Equal Weight Equity	2	0.0377
U.S. Mid Cap Value	2	0.0376
U.S. Equity	2	0.0375
Value Equity	2	0.0369
U.S. Revenue Weighted Equity	1	0.0188
Large Cap Blend Equities	1	0.0188
U.S. Mega Cap Value	1	0.0188
U.S. Size Factor Equity	1	0.0187
U.S. Large Cap Equity	1	0.0184
U.S. Buyback Equity	1	0.0183
Dividend Equity	1	0.0180
U.S. Quality / Factor Equity	1	0.0179
U.S. Moat Equity	1	0.0177
U.S. ESG Equity	1	0.0176

Global Equity 1 0.0168

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.005

ticker	weight_ols	name	category	etf_signal
VTV	1.021	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members

	ticker	weight	name	category	etf_signal
22	IWD	0.0192	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
2	RSP	0.0191	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
9	PRF	0.0191	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
29	VONV	0.0191	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
32	ILCV	0.0191	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
35	SCHV	0.0191	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
12	CFA	0.0190	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	LONG
18	FNDX	0.0190	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
24	IVE	0.0190	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
25	JPUS	0.0190	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	LONG
34	IUSV	0.0190	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
38	VTV	0.0190	Vanguard Value ETF	U.S. Large Cap Value	LONG
45	DTD	0.0190	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
50	DGRO	0.0190	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
15	NULV	0.0189	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
20	IWX	0.0189	iShares Russell Top 200 Value ETF	Value Equity	LONG
28	VOOV	0.0189	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
36	SPYV	0.0189	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
44	DLN	0.0189	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
5	FEX	0.0188	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
10	RWL	0.0188	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
30	MGV	0.0188	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
31	VYM	0.0188	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
39	IWS	0.0188	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
40	VOE	0.0188	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
42	PFM	0.0188	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
11	AIVL	0.0187	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
21	ONEV	0.0187	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	LONG
26	SIZE	0.0187	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
51	VIG	0.0187	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
53	JPME	0.0187	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	LONG

19	EQAL	0.0186	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	LONG
33	IMCB	0.0186	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
54	JVAL	0.0186	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
17	FDVV	0.0185	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
23	IWR	0.0185	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
37	VO	0.0185	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
47	DIA	0.0184	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
4	PKW	0.0183	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	FLAT
14	NOBL	0.0182	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
6	PWV	0.0181	Invesco Large Cap Value ETF	Value Equity	LONG
7	RDVY	0.0181	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	LONG
3	FVD	0.0180	First Trust VL Dividend	U.S. Dividend Equity	FLAT
16	SDY	0.0180	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
27	VLUE	0.0180	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
43	OUSA	0.0180	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
41	FTCS	0.0179	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	FLAT
49	LGLV	0.0179	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	FLAT
13	CFO	0.0178	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	LONG
8	MOAT	0.0177	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	LONG
48	USMV	0.0176	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
52	ETHO	0.0176	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	LONG
1	ACWV	0.0168	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
46	SPLV	0.0159	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT

Cluster 7 – IJR

Members: 48 **Var explained:** 0.880 **Signal:** LONG since 2026-04-10 (21 days, 2.9%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +35.3%)
- Latest sell_cross: 2026-02-27
- Current 252d return: +22.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-02-27)



Category breakdown

category	n_members	abs_weight
Financials	13	0.2576
U.S. Small Cap Value	6	0.1304
U.S. Small Cap Equity	5	0.1082
U.S. Mid Cap Value	3	0.0657
Dividend Equity	2	0.0424
Leveraged / Inverse	2	0.0422
U.S. Small Cap Low Volatility	2	0.0407
U.S. Small Cap Fundamental Equity	1	0.0218
Mid Cap Value Equities	1	0.0218
U.S. Mid Cap Dividend Equity	1	0.0217
Small Cap Value Equities	1	0.0216
U.S. Small Cap Dividend Equity	1	0.0216
U.S. Mid/Small Cap Equity	1	0.0216
U.S. Small/Mid Cap Fundamental Equity	1	0.0215
Value Equity	1	0.0215
U.S. Mid Cap Equity	1	0.0214
U.S. Small/Mid Cap Equity	1	0.0213
U.S. Equity	1	0.0210
U.S. Dividend Equity	1	0.0206
U.S. Mid Cap Low Volatility	1	0.0198
Global Equity	1	0.0181

Real Estate

1

0.0173

Replication basket – IJR (cor=0.98) basket lead at lag 1: -0.002

ticker	weight_ols	name	category	etf_signal
IJR	0.904	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
37	VBR	0.0220	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG
28	IJJ	0.0219	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
30	IVOV	0.0219	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
31	MDYV	0.0219	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	LONG
21	EZM	0.0218	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	LONG
23	FNDA	0.0218	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	LONG
8	FYX	0.0217	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	LONG
10	VIOO	0.0217	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	LONG
11	IJR	0.0217	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG
12	SPSM	0.0217	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	LONG
19	DON	0.0217	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	LONG
26	IWN	0.0217	iShares Russell 2000 Value ETF	U.S. Small Cap Value	LONG
29	IJS	0.0217	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
32	SLYV	0.0217	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	LONG
34	VTWV	0.0217	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	LONG
47	SDVY	0.0217	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
7	FNX	0.0216	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	LONG
18	DES	0.0216	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	LONG
20	EES	0.0216	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	LONG
33	VIOV	0.0216	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	LONG
9	PRFZ	0.0215	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	LONG
46	FYT	0.0215	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	LONG
41	RWK	0.0214	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	LONG
38	RWJ	0.0213	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	LONG
48	ISMD	0.0213	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
39	RFV	0.0212	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	LONG
42	OMFS	0.0210	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	LONG
13	FXO	0.0209	First Trust Financials AlphaDEX	Financials	FLAT
43	RZV	0.0209	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	LONG
6	REGL	0.0207	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
17	SMDV	0.0206	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	LONG

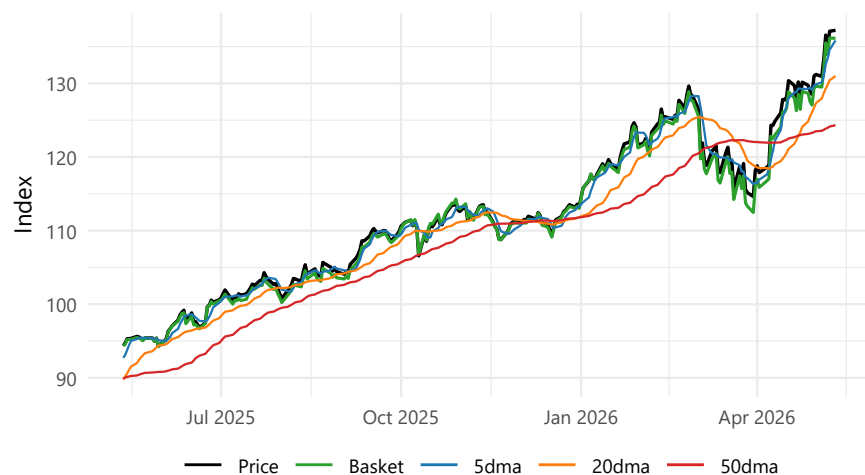
25	XSLV	0.0204	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	LONG
35	VFH	0.0204	Vanguard Financials ETF	Financials	FLAT
36	FNCL	0.0204	Fidelity MSCI Financials Index ETF	Financials	LONG
27	SMMV	0.0203	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
3	KBE	0.0202	State Street SPDR S&P Bank ETF	Financials	LONG
15	IYF	0.0202	iShares U.S. Financial ETF	Financials	FLAT
45	KCE	0.0202	State Street SPDR S&P Capital Markets E...	Financials	LONG
16	IYG	0.0201	iShares U.S. Financial Services ETF	Financials	FLAT
24	XMLV	0.0198	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
1	XLF	0.0197	State Street Financial Select Sector SP...	Financials	LONG
22	KBWB	0.0197	Invesco KBW Bank ETF	Financials	FLAT
2	KRE	0.0196	State Street SPDR S&P Regional Banking ...	Financials	LONG
14	IAT	0.0194	iShares U.S. Regional Banks ETF	Financials	LONG
44	IAI	0.0189	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
40	PSP	0.0181	Invesco Global Listed Private Equity ETF	Global Equity	LONG
4	KIE	0.0177	State Street SPDR S&P Insurance ETF	Financials	FLAT
5	REM	0.0173	iShares Mortgage Real Estate ETF	Real Estate	LONG

Cluster 8 – IEMG

Members: 25 **Var explained:** 0.875 **Signal:** LONG since 2026-04-10 (21 days, 9.9%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +49.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7724
Asia Equity	2	0.0812
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0357
Global Dividend Equity	1	0.0348
South Korea Equity	1	0.0341

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.023

ticker	weight_ols	name	category	etf_signal
IEMG	0.931	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members

	ticker	weight	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG

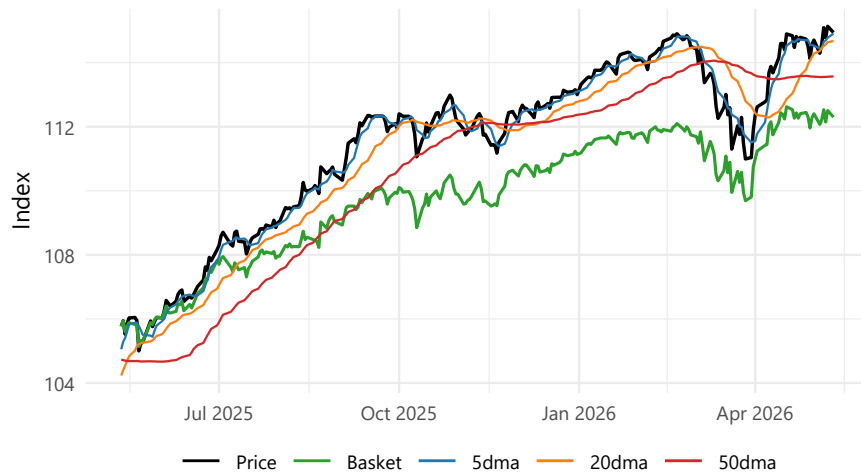
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
18	XSOE	0.0419	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0390	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0385	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0357	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0348	Global X SuperDividend ETF	Global Dividend Equity	LONG
25	FEMS	0.0347	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 9 – HYG

Members: 22 **Var explained:** 0.772 **Signal:** LONG since 2026-04-13 (20 days, 0.7%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +10.1%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +9.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-05)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.4918
Preferred Stock	7	0.2893
Emerging Markets Fixed Income	4	0.1695
Unknown	1	0.0493

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.048

ticker	weight_ols	name	category	etf_signal
HYG	0.911	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members

	ticker	weight	name	category	etf_signal
14	JNK	0.0502	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
1	HYG	0.0501	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
20	HYLB	0.0500	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
19	BBHY	0.0499	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG
17	SHYG	0.0497	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

15	SJNK	0.0496	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
8	PHB	0.0493			LONG
16	ANGL	0.0492	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
18	FALN	0.0492	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0488	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
10	PFF	0.0453	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0450	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0444	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
22	VWO	0.0443	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
12	PFXF	0.0438	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
21	PCY	0.0435	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0421	Invesco Preferred ETF	Preferred Stock	LONG
11	PSK	0.0416	State Street SPDR ICE Preferred Securit...	Preferred Stock	LONG
4	FPE	0.0410	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0407	Invesco Financial Preferred ETF	Preferred Stock	FLAT
13	HYEM	0.0373	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
9	VRP	0.0348	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 10 – SCHD

Members: 20 **Var explained:** 0.832 **Signal:** LONG since 2026-04-21 (14 days, 0.6%)

Signal logic:

- Latest buy_cross: 2026-04-21 – filter **PASS** (252d at date = +21.9%)
- Latest sell_cross: 2026-03-09
- Current 252d return: +20.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-21 > last sell at 2026-03-09)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	11	0.5763
MLP / Energy Infrastructure	4	0.1667
Value Equity	2	0.1037
Dividend Equity	1	0.0524
U.S. Defensive Equity	1	0.0515
U.S. Low Volatility Equity	1	0.0495

Replication basket – SCHD (cor=0.94) basket lead at lag 1: 0.022

ticker	weight_ols	name	category	etf_signal
SCHD	0.926	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members

	ticker	weight	name	category	etf_signal
15	DVY	0.0538	iShares Select Dividend ETF	U.S. Dividend Equity	LONG
20	SPYD	0.0535	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
14	SDOG	0.0534	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG

6	DHS	0.0533	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
19	ONEY	0.0528	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
11	FTA	0.0526	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	FLAT
5	FDL	0.0524	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
18	SPHD	0.0524	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
10	SCHD	0.0521	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
13	DIV	0.0517	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0516	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
7	CDC	0.0515	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	LONG
17	RPV	0.0511	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
12	PEY	0.0510	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	LONG
9	HDV	0.0508	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
8	LVHD	0.0495	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
1	EMLP	0.0472	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
4	MLPX	0.0419	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
2	AMLP	0.0393	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
3	MLPA	0.0383	Global X MLP ETF	MLP / Energy Infrastruct...	LONG

Cluster 11 – VHT

Members: 17 **Var explained:** 0.764 **Signal:** LONG since 2026-05-06 (3 days, -2.0%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +14.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.005

ticker	weight_ols	name	category	etf_signal
VHT	1.104	Vanguard Health Care ETF	Healthcare	FLAT

Members

	ticker	weight	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	FLAT
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	FLAT
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	FLAT
10	IXJ	0.0622	iShares Global Healthcare ETF	Healthcare	FLAT
1	XLV	0.0620	State Street Health Care Select Sector ...	Healthcare	FLAT

3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0573	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0545	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0542	VanEck Pharmaceutical ETF	Healthcare	FLAT
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0511	ARK Genomic Revolution ETF	Healthcare	LONG
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – URA + GUNR

Members: 17 **Var explained:** 0.756 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-08 – filter **PASS** (252d at date = +66.8%)
- Latest sell_cross: 2026-05-04
- Current 252d return: +54.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-04-08)



Category breakdown

category	n_members	abs_weight
Materials	13	0.7907
Consumer Staples	1	0.0589
Energy	1	0.0571
Norway Equity	1	0.0519
Industrial Metals	1	0.0414

Replication basket – URA+GUNR (cor=0.95) basket lead at lag 1: 0.035

ticker	weight_ols	name	category	etf_signal
URA	0.095	Global X Uranium ETF	Materials	LONG
GUNR	0.897	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members

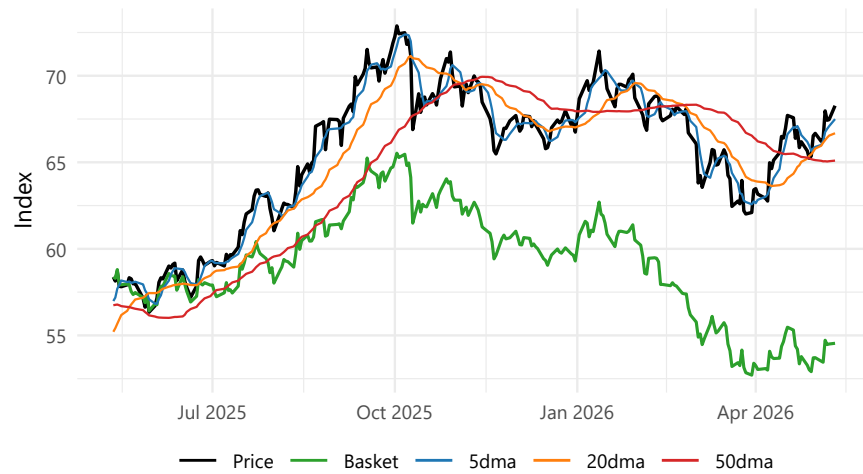
	ticker	weight	name	category	etf_signal
11	GNR	0.0653	State Street SPDR S&P Global Natural Re...	Materials	FLAT
6	MXI	0.0648	iShares Global Materials ETF	Materials	FLAT
10	GUNR	0.0643	FlexShares Global Upstream Natural Reso...	Materials	FLAT
7	IYM	0.0641	iShares U.S. Basic Materials ETF	Materials	FLAT
12	PICK	0.0627	iShares MSCI Global Select Metals & Min...	Materials	FLAT
9	VAW	0.0622	Vanguard Materials ETF	Materials	FLAT
17	NANR	0.0614	State Street SPDR S&P North American Na...	Materials	LONG
1	XLB	0.0612	State Street Materials Select Sector SP...	Materials	FLAT
8	RSPM	0.0611	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
13	SLX	0.0606	VanEck Steel ETF	Materials	LONG
4	XME	0.0605	State Street SPDR S&P Metals & Mining E...	Materials	LONG
5	MOO	0.0589	VanEck Agribusiness ETF	Consumer Staples	LONG
3	COPX	0.0587	Global X Copper Miners ETF	Materials	LONG
16	IGE	0.0571	iShares North American Natural Resource...	Energy	LONG
14	NORW	0.0519	Global X MSCI Norway ETF	Norway Equity	LONG
2	URA	0.0438	Global X Uranium ETF	Materials	LONG
15	DBB	0.0414	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 13 – KWEB

Members: 15 **Var explained:** 0.804 **Signal:** LONG since 2026-04-15 (18 days, 2.8%)

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-02-27
- Current 252d return: +20.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-15 > last sell at 2026-02-27)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0642
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	weight_ols	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members

	ticker	weight	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	LONG
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	LONG
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	LONG
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	LONG
3	EMQQ	0.0642	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
10	KBA	0.0594	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
15	KURE	0.0587	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – VNQ

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (21 days, 4.2%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-03-10
- Current 252d return: +13.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.001

ticker	weight_ols	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	LONG

Members

	ticker	weight	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG

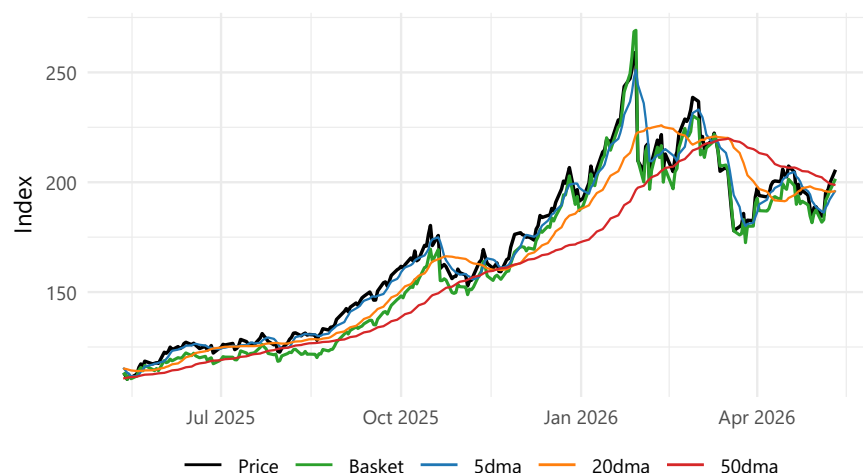
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 15 – GLD + IAU + SLV

Members: 13 **Var explained:** 0.831 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2025-11-11 – filter **PASS** (252d at date = +60.6%)
- Latest sell_cross: 2026-04-28
- Current 252d return: +79.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-28 more recent than last effective buy 2025-11-11)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.3889
Gold	3	0.2341
Precious Metals	2	0.1624
Silver	2	0.1525
Platinum	1	0.0622

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.017

ticker	weight_ols	name	category	etf_signal
GLD	0.115	SPDR Gold Shares	Gold	FLAT
IAU	0.623	iShares Gold Trust Shares	Gold	FLAT
SLV	0.418	iShares Silver Trust	Silver	LONG

Members

	ticker	weight	name	category	etf_signal
12	DBP	0.0814	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0810	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
2	GDXJ	0.0786	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0785	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0781	SPDR Gold Shares	Gold	FLAT
10	IAU	0.0781	iShares Gold Trust Shares	Gold	FLAT
11	SGOL	0.0779	abrdn Physical Gold Shares ETF	Gold	FLAT
13	GOAU	0.0777	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0776	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0765	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0763	iShares Silver Trust	Silver	LONG
9	SIVR	0.0761	abrdn Physical Silver Shares ETF	Silver	LONG
8	PPLT	0.0622	abrdn Physical Platinum Shares ETF	Platinum	LONG

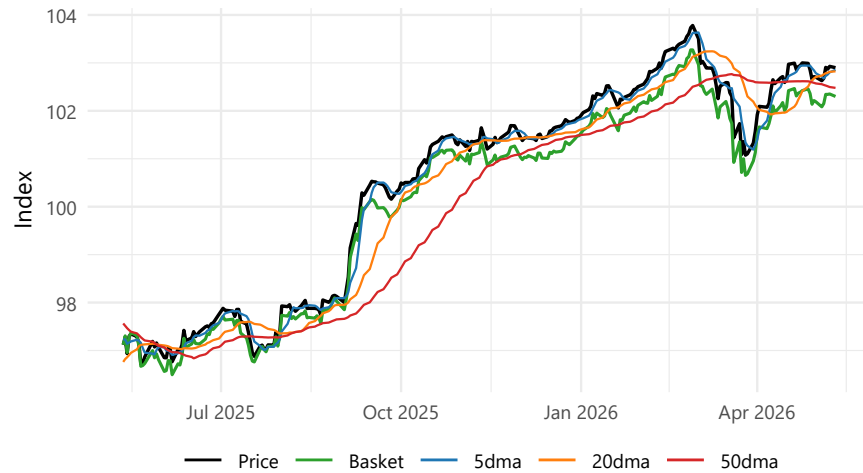
Cluster 16 – MUB

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-14 – filter **PASS** (252d at date = +6.9%)
- Latest sell_cross: 2026-05-05

- Current 252d return: +5.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-05 more recent than last effective buy 2026-04-14)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.181

ticker	weight_ols	name	category	etf_signal
MUB	0.905	iShares National Muni Bond ETF	Municipal Bond	LONG

Members

	ticker	weight	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG

1	FMB	0.0796	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0668	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – XLU + XLP

Members: 11 **Var explained:** 0.748 **Signal:** LONG since 2026-05-01 (6 days, -2.3%)

Signal logic:

- Latest buy_cross: 2026-05-01 – filter **PASS** (252d at date = +9.8%)
- Latest sell_cross: 2026-03-09
- Current 252d return: +7.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-01 > last sell at 2026-03-09)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5726
Utilities	4	0.3528
U.S. Small Cap Consumer Staples	1	0.0746

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.007

ticker	weight_ols	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members

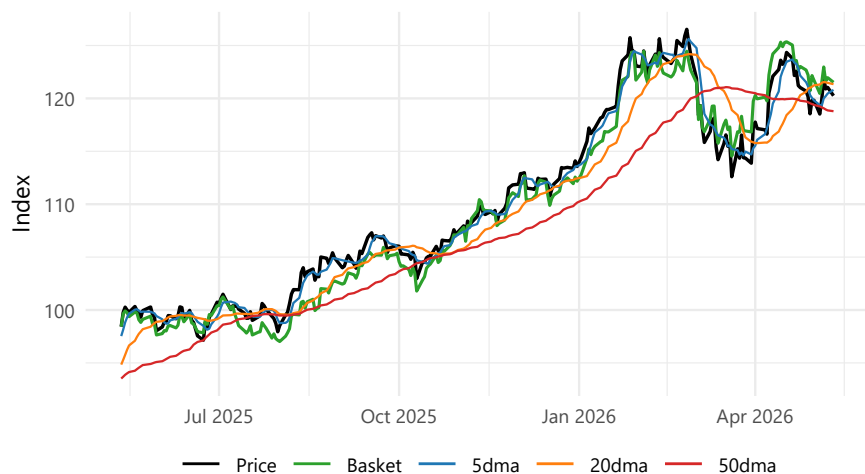
	ticker	weight	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0973	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0947	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0944	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0892	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0880	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0880	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0875	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0746	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	FLAT

Cluster 18 – EWZ + ILF + DBA + ECH

Members: 11 **Var explained:** 0.325 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +36.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +23.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Latin America Equity	1	0.1565
Brazil Equity	1	0.1432
Chile Equity	1	0.1273
Malaysia Equity	1	0.1242
Indonesia Equity	1	0.1137
Philippines Equity	1	0.1051
Vietnam Equity	1	0.0792
Agricultural Commodities	1	0.0598
Turkey Equity	1	0.0575
Unknown	1	0.0189
Natural Gas	1	0.0147

Replication basket – EWZ+ILF+DBA+ECH (cor=0.93) basket lead at lag 1: 0.035

ticker	weight_ols	name	category	etf_signal
EWZ	0.047	iShares MSCI Brazil ETF	Brazil Equity	FLAT
ILF	0.382	iShares Latin America 40 ETF	Latin America Equity	FLAT
DBA	0.056	Invesco DB Agriculture Fund	Agricultural Commodit...	FLAT
ECH	0.158	iShares MSCI Chile ETF	Chile Equity	LONG

Members

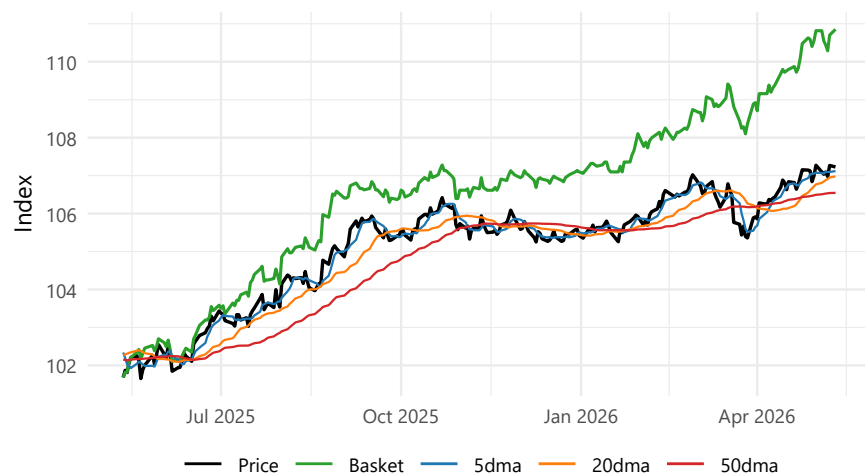
	ticker	weight	name	category	etf_signal
4	ILF	0.1565	iShares Latin America 40 ETF	Latin America Equity	FLAT
1	EWZ	0.1432	iShares MSCI Brazil ETF	Brazil Equity	FLAT
3	ECH	0.1273	iShares MSCI Chile ETF	Chile Equity	LONG
8	EWM	0.1242	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1137	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1051	iShares MSCI Philippines ETF	Philippines Equity	FLAT
11	VNM	0.0792	VanEck Vietnam ETF	Vietnam Equity	LONG
2	DBA	0.0598	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
10	TUR	0.0575	iShares MSCI Turkey ETF	Turkey Equity	LONG
9	RSX	0.0189			LONG
5	UNG	0.0147	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 19 – VTIP

Members: 10 **Var explained:** 0.908 **Signal:** LONG since 2026-04-09 (22 days, 0.8%)

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +4.1%)
- Latest sell_cross: 2026-03-20
- Current 252d return: +4.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-09 > last sell at 2026-03-20)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP (cor=0.90) basket lead at lag 1: -0.010

ticker	weight_ols	name	category	etf_signal
VTIP	1.724	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT

Members

	ticker	weight	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	LONG
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	LONG
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	LONG
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	LONG

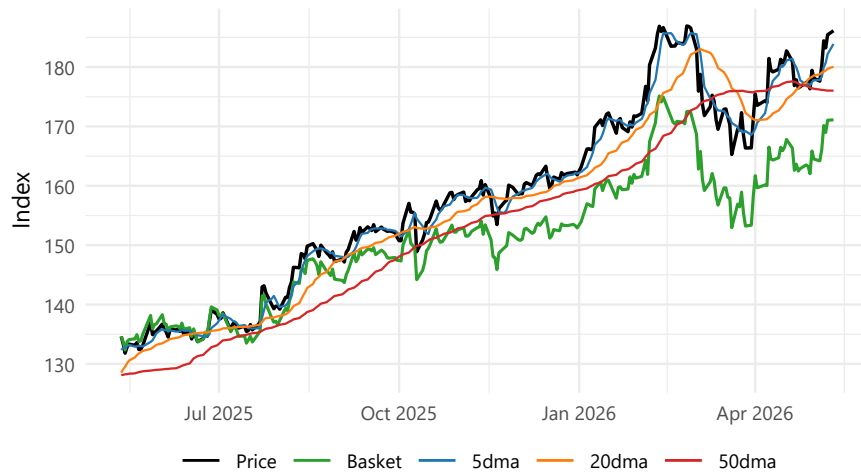
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG

Cluster 20 – EWJ

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +41.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7796
Asia-Pacific Equity	2	0.2204

Replication basket – EWJ (cor=0.96) basket lead at lag 1: -0.037

ticker	weight_ols	name	category	etf_signal
EWJ	0.908	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members

	ticker	weight	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
9	HEWJ	0.1115	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1114	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1106	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1091	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 21 – PDBC

Members: 9 **Var explained:** 0.913 **Signal:** LONG since 2026-01-08 (84 days, 47.8%)

Signal logic:

- Latest buy_cross: 2026-01-08 – filter **PASS** (252d at date = +2.7%)
- Latest sell_cross: 2025-12-15
- Current 252d return: +61.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-01-08 > last sell at 2025-12-15)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183
Energy Commodities	1	0.1119

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.070

ticker	weight_ols	name	category	etf_signal
PDBC	1.173	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members

	ticker	weight	name	category	etf_signal
5	GSG	0.1151	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1149	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
7	PDBC	0.1149	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
8	COMT	0.1148	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1098	United States Oil Fund	Oil	LONG

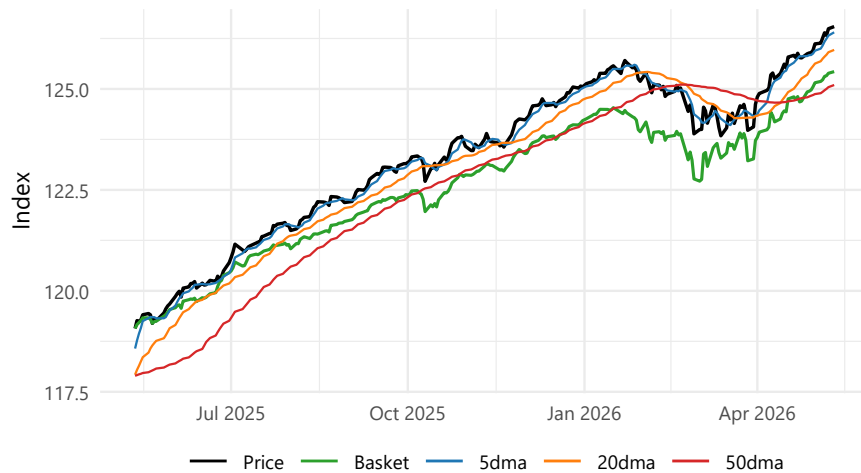
4	DBO	0.1085	Invesco DB Oil Fund	Oil	FLAT
6	DJP	0.1051	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	FTGC	0.1050	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 22 – FLOT + BKLN + SRLN + FLRN + FLTR

Members: 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-07 (24 days, 1.2%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-02-03
- Current 252d return: +6.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3632
Interest Rate Hedged Fixed Income	3	0.3245

Floating Rate Fixed Income 3 0.3123

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

ticker	weight_ols	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.209	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.311	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members

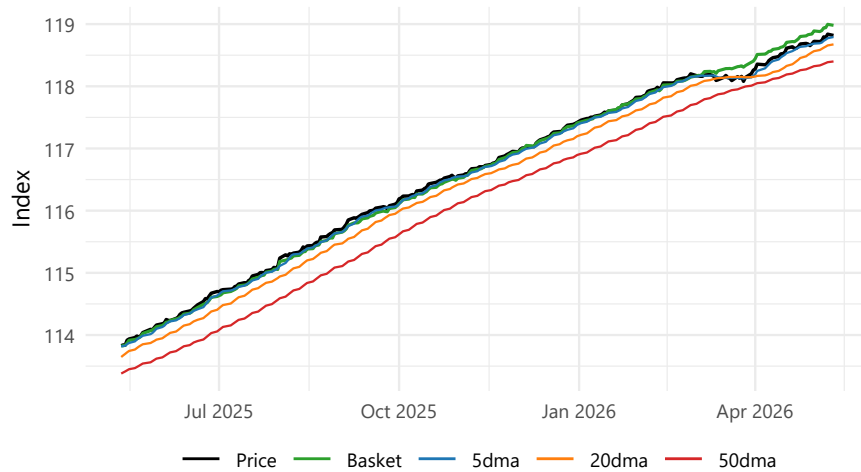
	ticker	weight	name	category	etf_signal
2	SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1213	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1170	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1081	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1035	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1008	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 23 – SHV + MINT + GBIL + ICSH

Members: 9 **Var explained:** 0.457 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2022-10-31 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-03-24
- Current 252d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-24 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.108

ticker	weight_ols	name	category	etf_signal
SHV	0.350	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.175	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.117	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.418	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members

	ticker	weight	name	category	etf_signal
5	GSY	0.1359	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1286	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1261	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1244	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1090	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT

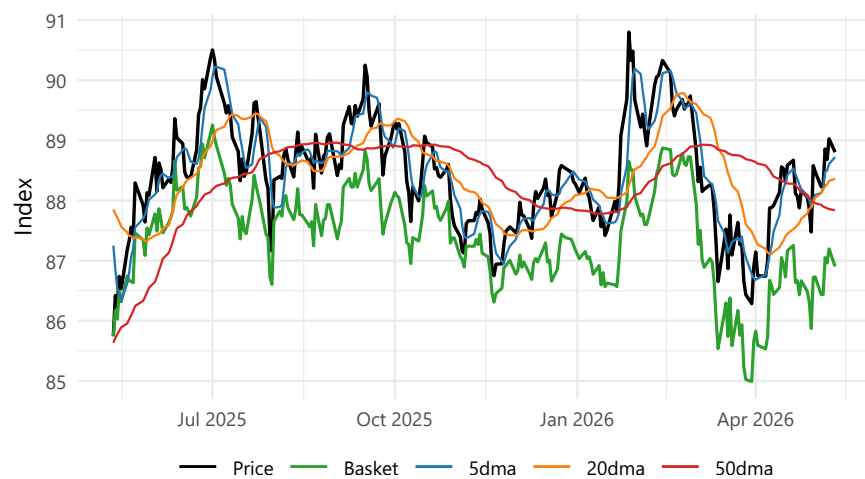
2	MINT	0.1062	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.1032	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0879	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0787	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 24 – BWX

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-04-30 (7 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-04-30 – filter **PASS** (252d at date = +0.3%)
- Latest sell_cross: 2026-03-03
- Current 252d return: +2.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-30 > last sell at 2026-03-03)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641

International Fixed Income	2	0.359
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Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.029

ticker	weight_ols	name	category	etf_signal
BWX	0.752	State Street SPDR Bloomberg Internati...	International Fixed I...	LONG

Members

	ticker	weight	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	LONG
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	LONG
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1647	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXV	0.1453	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 25 – INDA + INDY + EPI + PIN

Members: 4 **Var explained:** 0.964 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-07 – filter **BLOCKED** (252d at date = -9.2%; needs > 0)
- Latest effective buy: 2026-02-06 – delayed re-arm from blocked cross (252d at fire = +2.9%)
- Latest sell_cross: 2026-05-04
- Current 252d return: -7.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-02-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.7508
Unknown	1	0.2492

Replication basket – INDIA+INDY+EPI+PIN (cor=1.00) basket lead at lag 1: -0.071

ticker	weight_ols	name	category	etf_signal
INDA	0.250	iShares MSCI India ETF	India Equity	FLAT
INDY	0.246	iShares India 50 ETF	India Equity	FLAT
EPI	0.256	WisdomTree India Earnings Fund	India Equity	LONG
PIN	0.249			FLAT

Members

	ticker	weight	name	category	etf_signal
1	INDA	0.2527	iShares MSCI India ETF	India Equity	FLAT
3	EPI	0.2493	WisdomTree India Earnings Fund	India Equity	LONG
4	PIN	0.2492			FLAT

2 INDY 0.2489 iShares India 50 ETF India Equity FLAT

Cluster 26 – SQQQ + UUP

Members: 2 Var explained: 0.624 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-04-08
- Current 252d return: -45.4% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)



Category breakdown

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.017

ticker	weight_ols	name	category	etf_signal
SQQQ	0.507	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.496	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members

ticker	weight	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
MINT	cluster_23	SHV + MINT + GBIL + ICSH	2023-01-30	822	18.7%	0.00	1.00
GBIL	cluster_23	SHV + MINT + GBIL + ICSH	2022-08-18	934	17.5%	0.00	1.00
BIL	cluster_23	SHV + MINT + GBIL + ICSH	2022-05-19	996	18.0%	0.00	1.00
GSY	cluster_23	SHV + MINT + GBIL + ICSH	2022-12-29	842	19.5%	0.00	1.00
VNLA	cluster_23	SHV + MINT + GBIL + ICSH	2026-04-02	26	0.5%	0.03	0.97
SRLN	cluster_22	FLOT + BKLN + SRLN + FL...	2026-04-07	24	1.9%	0.00	1.00
FTSL	cluster_22	FLOT + BKLN + SRLN + FL...	2026-04-06	25	1.4%	0.00	1.00
BKLN	cluster_22	FLOT + BKLN + SRLN + FL...	2026-04-06	25	1.3%	0.00	1.00
LQDH	cluster_22	FLOT + BKLN + SRLN + FL...	2026-04-06	25	1.3%	0.01	0.99
IGBH	cluster_22	FLOT + BKLN + SRLN + FL...	2026-04-08	23	1.7%	0.01	0.99
IGHG	cluster_22	FLOT + BKLN + SRLN + FL...	2026-04-07	24	1.1%	0.11	0.89
NXTG	cluster_3	QQQ	2026-04-08	23	21.7%	0.00	1.00
QTEC	cluster_3	QQQ	2026-04-10	21	31.6%	0.00	1.00
XSD	cluster_3	QQQ	2026-04-09	22	58.0%	0.00	1.00
PTNQ	cluster_3	QQQ	2026-04-17	16	9.9%	0.00	1.00
FTXL	cluster_3	QQQ	2026-04-07	24	59.1%	0.00	1.00
PSI	cluster_3	QQQ	2026-04-07	24	56.9%	0.00	1.00
SOXX	cluster_3	QQQ	2026-04-08	23	43.8%	0.00	1.00
SMH	cluster_3	QQQ	2026-04-08	23	36.3%	0.00	1.00
AIQ	cluster_3	QQQ	2026-04-13	20	24.3%	0.00	1.00
IYW	cluster_3	QQQ	2026-04-10	21	23.6%	0.00	1.00
XNTK	cluster_3	QQQ	2026-04-10	21	24.0%	0.00	1.00
XLK	cluster_3	QQQ	2026-04-10	21	24.7%	0.00	1.00
IXN	cluster_3	QQQ	2026-04-10	21	24.5%	0.00	1.00
PSCT	cluster_3	QQQ	2026-04-07	24	30.0%	0.00	1.00
QQQ	cluster_3	QQQ	2026-04-10	21	16.7%	0.00	1.00
QTUM	cluster_3	QQQ	2026-04-10	21	26.7%	0.00	1.00
PTLC	cluster_3	QQQ	2026-04-20	15	4.3%	0.00	1.00
FTEC	cluster_3	QQQ	2026-04-10	21	22.5%	0.00	1.00
VGT	cluster_3	QQQ	2026-04-10	21	22.4%	0.00	1.00
CIBR	cluster_3	QQQ	2026-04-17	16	13.5%	0.00	1.00
QYLD	cluster_3	QQQ	2026-04-08	23	4.3%	0.00	1.00
IGM	cluster_3	QQQ	2026-04-09	22	22.2%	0.00	1.00
PTF	cluster_3	QQQ	2026-04-07	24	35.1%	0.00	1.00
BOTZ	cluster_3	QQQ	2026-04-16	17	12.5%	0.00	1.00
SKYY	cluster_3	QQQ	2026-04-17	16	9.4%	0.00	1.00
FXL	cluster_3	QQQ	2026-04-09	22	18.0%	0.00	1.00

XT	cluster_3	QQQ	2026-04-13	20	10.8%	0.00	1.00
TDIV	cluster_3	QQQ	2026-04-13	20	14.3%	0.00	1.00
ROBO	cluster_3	QQQ	2026-04-14	19	14.2%	0.00	1.00
ONEQ	cluster_3	QQQ	2026-04-13	20	13.4%	0.00	1.00
HACK	cluster_3	QQQ	2026-04-17	16	8.6%	0.00	1.00
ICVT	cluster_3	QQQ	2026-04-08	23	10.4%	0.00	1.00
QQEW	cluster_3	QQQ	2026-04-16	17	6.9%	0.00	1.00
PWB	cluster_3	QQQ	2026-04-10	21	13.7%	0.00	1.00
SPHB	cluster_3	QQQ	2026-04-10	21	16.4%	0.00	1.00
CWB	cluster_3	QQQ	2026-04-08	23	10.1%	0.00	1.00
NULG	cluster_3	QQQ	2026-04-10	21	13.7%	0.00	1.00
IVW	cluster_3	QQQ	2026-04-13	20	11.5%	0.00	1.00
SPYG	cluster_3	QQQ	2026-04-13	20	11.5%	0.00	1.00
VOOG	cluster_3	QQQ	2026-04-13	20	11.5%	0.00	1.00
IUSG	cluster_3	QQQ	2026-04-13	20	11.2%	0.00	1.00
MGK	cluster_3	QQQ	2026-04-13	20	11.9%	0.00	1.00
ILCG	cluster_3	QQQ	2026-04-13	20	11.5%	0.00	1.00
VUG	cluster_3	QQQ	2026-04-13	20	11.4%	0.00	1.00
BLCN	cluster_3	QQQ	2026-04-15	18	6.0%	0.00	1.00
IGV	cluster_3	QQQ	2026-04-21	14	4.6%	0.00	1.00
BLOK	cluster_3	QQQ	2026-04-13	20	17.0%	0.00	1.00
SCHG	cluster_3	QQQ	2026-04-13	20	10.3%	0.00	1.00
ARKW	cluster_3	QQQ	2026-04-14	19	13.7%	0.01	0.99
PRNT	cluster_3	QQQ	2026-04-16	17	6.6%	0.01	0.99
IWY	cluster_3	QQQ	2026-04-14	19	7.7%	0.01	0.99
VONG	cluster_3	QQQ	2026-04-14	19	7.3%	0.01	0.99
IWF	cluster_3	QQQ	2026-04-14	19	7.3%	0.02	0.98
ARKQ	cluster_3	QQQ	2026-04-14	19	11.1%	0.06	0.94
IPO	cluster_3	QQQ	2026-04-13	20	7.6%	0.09	0.91
FDN	cluster_3	QQQ	2026-04-10	21	8.6%	0.18	0.82
ARKK	cluster_3	QQQ	2026-04-14	19	6.9%	0.23	0.77
FINX	cluster_3	QQQ	2026-04-20	15	-3.4%	0.44	0.56
PNQI	cluster_3	QQQ	2026-04-15	18	-1.6%	0.55	0.45
PBS	cluster_3	QQQ	2026-04-14	19			
EWT	cluster_8	IEMG	2026-04-08	23	27.2%	0.00	1.00
EWY	cluster_8	IEMG	2026-04-09	22	38.5%	0.00	1.00
AIA	cluster_8	IEMG	2026-04-10	21	21.0%	0.00	1.00
AAXJ	cluster_8	IEMG	2026-04-13	20	13.7%	0.00	1.00
EEMA	cluster_8	IEMG	2026-04-13	20	13.4%	0.00	1.00
EMGF	cluster_8	IEMG	2026-04-10	21	13.3%	0.00	1.00
XSOE	cluster_8	IEMG	2026-04-10	21	12.5%	0.00	1.00
EEM	cluster_8	IEMG	2026-04-10	21	12.1%	0.00	1.00
IEMG	cluster_8	IEMG	2026-04-10	21	11.7%	0.00	1.00
ESGE	cluster_8	IEMG	2026-04-13	20	11.2%	0.00	1.00
GEM	cluster_8	IEMG	2026-04-10	21	11.3%	0.00	1.00

GMF	cluster_8	IEMG	2026-04-14	19	6.1%	0.00	1.00
DEM	cluster_8	IEMG	2026-04-09	22	7.0%	0.00	1.00
EEMV	cluster_8	IEMG	2026-04-10	21	8.5%	0.00	1.00
PXH	cluster_8	IEMG	2026-04-10	21	5.7%	0.01	0.99
SPEM	cluster_8	IEMG	2026-04-13	20	5.7%	0.02	0.98
VWO	cluster_8	IEMG	2026-04-13	20	5.6%	0.02	0.98
SCHE	cluster_8	IEMG	2026-04-13	20	5.2%	0.03	0.97
EWX	cluster_8	IEMG	2026-04-10	21	8.5%	0.03	0.97
FNDE	cluster_8	IEMG	2026-04-10	21	4.9%	0.03	0.97
FEM	cluster_8	IEMG	2026-04-07	24	11.5%	0.09	0.91
DGS	cluster_8	IEMG	2026-04-10	21	4.0%	0.15	0.85
SDIV	cluster_8	IEMG	2026-04-09	22	1.2%	0.46	0.54
EIS	cluster_4	V00	2026-04-10	21	8.4%	0.00	1.00
DSI	cluster_4	V00	2026-04-14	19	8.6%	0.00	1.00
IOO	cluster_4	V00	2026-04-13	20	9.1%	0.00	1.00
QAI	cluster_4	V00	2026-04-08	23	4.0%	0.00	1.00
OEF	cluster_4	V00	2026-04-13	20	9.5%	0.00	1.00
SUSA	cluster_4	V00	2026-04-13	20	8.0%	0.00	1.00
OMFL	cluster_4	V00	2026-04-10	21	7.2%	0.00	1.00
ESGU	cluster_4	V00	2026-04-13	20	8.1%	0.00	1.00
MGC	cluster_4	V00	2026-04-13	20	8.7%	0.00	1.00
IVV	cluster_4	V00	2026-04-13	20	7.7%	0.00	1.00
SPY	cluster_4	V00	2026-04-13	20	7.8%	0.00	1.00
VOO	cluster_4	V00	2026-04-13	20	7.7%	0.00	1.00
XLG	cluster_4	V00	2026-04-13	20	9.4%	0.00	1.00
VV	cluster_4	V00	2026-04-13	20	7.7%	0.00	1.00
IWL	cluster_4	V00	2026-04-13	20	8.2%	0.00	1.00
ILCB	cluster_4	V00	2026-04-13	20	7.7%	0.00	1.00
SPTM	cluster_4	V00	2026-04-13	20	7.4%	0.00	1.00
IYY	cluster_4	V00	2026-04-13	20	7.5%	0.00	1.00
EPS	cluster_4	V00	2026-04-10	21	8.1%	0.00	1.00
SCHB	cluster_4	V00	2026-04-13	20	7.4%	0.00	1.00
ITOT	cluster_4	V00	2026-04-13	20	7.4%	0.00	1.00
SCHX	cluster_4	V00	2026-04-13	20	7.5%	0.00	1.00
VTI	cluster_4	V00	2026-04-13	20	7.4%	0.00	1.00
SPYX	cluster_4	V00	2026-04-13	20	7.8%	0.00	1.00
IWB	cluster_4	V00	2026-04-13	20	7.2%	0.00	1.00
SPHQ	cluster_4	V00	2026-04-10	21	5.4%	0.00	1.00
IWV	cluster_4	V00	2026-04-13	20	7.2%	0.00	1.00
VTHR	cluster_4	V00	2026-04-13	20	7.3%	0.00	1.00
VONE	cluster_4	V00	2026-04-13	20	7.2%	0.00	1.00
CATH	cluster_4	V00	2026-04-13	20	7.3%	0.00	1.00
TILT	cluster_4	V00	2026-04-13	20	6.0%	0.00	1.00
ACWI	cluster_4	V00	2026-04-13	20	6.2%	0.01	0.99
JHML	cluster_4	V00	2026-04-13	20	5.6%	0.01	0.99

SPGM	cluster_4	V00	2026-04-10	21	7.0%	0.01	0.99
VT	cluster_4	V00	2026-04-13	20	6.1%	0.01	0.99
LRGF	cluster_4	V00	2026-04-13	20	6.4%	0.01	0.99
QDF	cluster_4	V00	2026-04-13	20	5.3%	0.01	0.99
CRBN	cluster_4	V00	2026-04-13	20	6.2%	0.01	0.99
AOA	cluster_4	V00	2026-04-13	20	4.8%	0.01	0.99
GSLC	cluster_4	V00	2026-04-13	20	6.2%	0.01	0.99
URTH	cluster_4	V00	2026-04-13	20	5.7%	0.01	0.99
AOR	cluster_4	V00	2026-04-13	20	3.6%	0.02	0.98
AOM	cluster_4	V00	2026-04-13	20	2.4%	0.05	0.95
QUS	cluster_4	V00	2026-04-14	19	2.5%	0.06	0.94
FPXI	cluster_4	V00	2026-04-10	21	7.3%	0.07	0.93
AOK	cluster_4	V00	2026-04-13	20	1.8%	0.07	0.93
QUAL	cluster_4	V00	2026-04-14	19	3.6%	0.08	0.92
DGRW	cluster_4	V00	2026-04-14	19	3.2%	0.11	0.89
XLY	cluster_4	V00	2026-04-15	18	1.0%	0.32	0.68
FDIS	cluster_4	V00	2026-04-14	19	2.4%	0.34	0.66
VCR	cluster_4	V00	2026-04-14	19	2.4%	0.35	0.65
VOX	cluster_4	V00	2026-04-13	20	2.3%	0.60	0.40
FCOM	cluster_4	V00	2026-04-13	20	2.1%	0.62	0.38
XLC	cluster_4	V00	2026-04-14	19	-0.8%	0.74	0.24
IYC	cluster_4	V00	2026-04-14	19	-1.0%	0.82	0.18
EWS	cluster_4	V00	2026-04-07	24	1.9%	0.83	0.16
RTH	cluster_4	V00	2026-04-10	21	-0.0%	0.93	0.07
SPLG	cluster_4	V00	2026-04-13	20			
VPL	cluster_20	EWJ	2026-04-10	21	10.1%	0.00	1.00
EWJ	cluster_20	EWJ	2026-04-13	20	4.4%	0.03	0.97
BBJP	cluster_20	EWJ	2026-04-30	7	3.4%	0.03	0.97
HEWJ	cluster_20	EWJ	2026-04-08	23	2.7%	0.09	0.91
DBJP	cluster_20	EWJ	2026-04-08	23	2.5%	0.13	0.87
EWJV	cluster_20	EWJ	2026-05-06	3	-0.1%	0.17	0.83
DXJ	cluster_20	EWJ	2026-05-01	6	4.2%	0.23	0.77
VLUE	cluster_6	VTV	2026-04-09	22	21.2%	0.00	1.00
FNDX	cluster_6	VTV	2026-04-09	22	5.6%	0.00	1.00
PRF	cluster_6	VTV	2026-04-09	22	6.4%	0.00	1.00
IWX	cluster_6	VTV	2026-04-10	21	6.3%	0.00	1.00
NULV	cluster_6	VTV	2026-04-10	21	6.7%	0.00	1.00
JVAL	cluster_6	VTV	2026-04-10	21	9.1%	0.00	1.00
IWD	cluster_6	VTV	2026-04-10	21	5.8%	0.00	1.00
VONV	cluster_6	VTV	2026-04-10	21	5.8%	0.00	1.00
MGV	cluster_6	VTV	2026-04-13	20	3.5%	0.00	1.00
ILCV	cluster_6	VTV	2026-04-13	20	4.0%	0.00	1.00
VOOV	cluster_6	VTV	2026-04-14	19	3.3%	0.00	1.00
SPYV	cluster_6	VTV	2026-04-14	19	3.2%	0.00	1.00
IVE	cluster_6	VTV	2026-04-14	19	3.2%	0.00	1.00

SCHV	cluster_6	VTV	2026-04-10	21	4.4%	0.00	1.00
IUSV	cluster_6	VTV	2026-04-14	19	3.1%	0.00	1.00
RWL	cluster_6	VTV	2026-04-10	21	4.8%	0.01	0.99
VTV	cluster_6	VTV	2026-04-13	20	2.8%	0.01	0.99
IMCB	cluster_6	VTV	2026-04-10	21	5.5%	0.04	0.96
FEX	cluster_6	VTV	2026-04-09	22	4.2%	0.06	0.94
IWS	cluster_6	VTV	2026-04-09	22	4.2%	0.06	0.94
IWR	cluster_6	VTV	2026-04-10	21	4.9%	0.08	0.92
ETHO	cluster_6	VTV	2026-04-09	22	5.6%	0.11	0.89
VO	cluster_6	VTV	2026-04-13	20	3.3%	0.11	0.89
SIZE	cluster_6	VTV	2026-04-13	20	3.4%	0.12	0.88
DTD	cluster_6	VTV	2026-04-13	20	2.5%	0.12	0.88
AIVL	cluster_6	VTV	2026-04-10	21	3.2%	0.12	0.88
DLN	cluster_6	VTV	2026-04-13	20	2.6%	0.12	0.88
RSP	cluster_6	VTV	2026-04-14	19	2.2%	0.20	0.80
VIG	cluster_6	VTV	2026-04-14	19	2.0%	0.25	0.75
PFM	cluster_6	VTV	2026-04-14	19	1.9%	0.26	0.74
FDVV	cluster_6	VTV	2026-04-14	19	2.4%	0.27	0.73
DIA	cluster_6	VTV	2026-04-14	19	2.4%	0.29	0.71
JPME	cluster_6	VTV	2026-04-07	24	3.9%	0.34	0.66
VYM	cluster_6	VTV	2026-04-10	21	2.5%	0.38	0.62
OUSA	cluster_6	VTV	2026-04-20	15	-0.9%	0.47	0.53
DGRO	cluster_6	VTV	2026-04-13	20	1.2%	0.54	0.46
MOAT	cluster_6	VTV	2026-05-06	3	-0.9%	0.61	0.38
PWV	cluster_6	VTV	2026-04-02	26	3.6%	0.62	0.38
RDVY	cluster_6	VTV	2026-04-10	21	3.0%	0.70	0.30
CFO	cluster_6	VTV	2026-04-13	20	0.7%	0.72	0.28
SDY	cluster_6	VTV	2026-05-01	6	-0.9%	0.71	0.28
CFA	cluster_6	VTV	2026-04-13	20	0.7%	0.73	0.27
USMV	cluster_6	VTV	2026-05-05	4	-0.4%	0.64	0.24
EQAL	cluster_6	VTV	2026-04-08	23	1.5%	0.76	0.24
ACWV	cluster_6	VTV	2026-05-06	3	-0.4%	0.66	0.21
ONEV	cluster_6	VTV	2026-04-14	19	0.3%	0.85	0.15
JPUS	cluster_6	VTV	2026-04-08	23	0.9%	0.85	0.15
VOE	cluster_6	VTV	2026-04-09	22	1.1%	0.85	0.15
PDBC	cluster_21	PDBC	2025-12-22	95	39.8%	0.01	0.99
DBC	cluster_21	PDBC	2025-12-22	95	38.7%	0.02	0.98
USO	cluster_21	PDBC	2026-01-29	70	75.2%	0.30	0.68
FXF	cluster_24	BWX	2026-05-05	4	0.6%	0.12	0.88
FXB	cluster_24	BWX	2026-04-13	20	1.0%	0.13	0.87
IGOV	cluster_24	BWX	2026-05-04	5	1.1%	0.19	0.81
BWX	cluster_24	BWX	2026-05-11	0	0.0%	0.35	0.65
CQQQ	cluster_13	KWEB	2026-04-20	15	8.0%	0.00	1.00
CNYA	cluster_13	KWEB	2026-04-14	19	5.8%	0.00	1.00
ASHR	cluster_13	KWEB	2026-04-13	20	6.9%	0.01	0.99

EWI	cluster_13	KWEB	2026-04-08	23	2.3%	0.01	0.99
KBA	cluster_13	KWEB	2026-04-10	21	9.8%	0.01	0.99
CNXT	cluster_13	KWEB	2026-04-10	21	17.6%	0.01	0.99
CXSE	cluster_13	KWEB	2026-04-14	19	3.7%	0.17	0.83
GXC	cluster_13	KWEB	2026-05-04	5	2.0%	0.19	0.81
FXI	cluster_13	KWEB	2026-04-30	7	1.8%	0.28	0.72
MCHI	cluster_13	KWEB	2026-05-06	3	-0.0%	0.36	0.63
KGRN	cluster_13	KWEB	2026-05-06	3	1.6%	0.62	0.04
FV	cluster_2	IJH	2026-04-14	19	8.9%	0.00	1.00
MTUM	cluster_2	IJH	2026-04-08	23	18.1%	0.00	1.00
PRN	cluster_2	IJH	2026-04-08	23	15.4%	0.00	1.00
FPX	cluster_2	IJH	2026-04-08	23	12.4%	0.00	1.00
RPG	cluster_2	IJH	2026-04-09	22	14.2%	0.00	1.00
PDP	cluster_2	IJH	2026-04-08	23	9.6%	0.00	1.00
VFMO	cluster_2	IJH	2026-04-08	23	10.5%	0.00	1.00
XMMO	cluster_2	IJH	2026-04-06	25	13.4%	0.00	1.00
SCHA	cluster_2	IJH	2026-04-09	22	8.6%	0.00	1.00
FTC	cluster_2	IJH	2026-04-10	21	9.2%	0.00	1.00
IMCG	cluster_2	IJH	2026-04-09	22	9.1%	0.00	1.00
SCHM	cluster_2	IJH	2026-04-09	22	7.2%	0.01	0.99
PSC	cluster_2	IJH	2026-04-10	21	8.0%	0.01	0.99
IWM	cluster_2	IJH	2026-04-10	21	9.2%	0.01	0.99
VTWO	cluster_2	IJH	2026-04-10	21	9.2%	0.01	0.99
IWO	cluster_2	IJH	2026-04-13	20	10.0%	0.01	0.99
VTWG	cluster_2	IJH	2026-04-13	20	10.0%	0.01	0.99
IWC	cluster_2	IJH	2026-04-09	22	10.0%	0.01	0.99
IJK	cluster_2	IJH	2026-04-09	22	6.9%	0.01	0.99
MDYG	cluster_2	IJH	2026-04-09	22	6.9%	0.01	0.99
IVOG	cluster_2	IJH	2026-04-09	22	6.8%	0.02	0.98
VOT	cluster_2	IJH	2026-04-14	19	4.9%	0.02	0.98
VIOG	cluster_2	IJH	2026-04-08	23	6.6%	0.02	0.98
QMOM	cluster_2	IJH	2026-04-08	23	9.1%	0.02	0.98
SLYG	cluster_2	IJH	2026-04-08	23	6.6%	0.02	0.98
IJT	cluster_2	IJH	2026-04-08	23	6.6%	0.02	0.98
PAVE	cluster_2	IJH	2026-04-10	21	4.4%	0.02	0.98
XSMO	cluster_2	IJH	2026-04-08	23	7.6%	0.02	0.98
VBK	cluster_2	IJH	2026-04-10	21	9.9%	0.03	0.97
XMHQ	cluster_2	IJH	2026-04-10	21	3.6%	0.03	0.97
ISCG	cluster_2	IJH	2026-04-10	21	7.4%	0.05	0.95
DWAS	cluster_2	IJH	2026-04-07	24	13.7%	0.06	0.94
SMLF	cluster_2	IJH	2026-04-09	22	6.4%	0.06	0.94
VXF	cluster_2	IJH	2026-04-10	21	7.2%	0.07	0.93
IVOO	cluster_2	IJH	2026-04-10	21	4.8%	0.07	0.93
SPMD	cluster_2	IJH	2026-04-10	21	4.8%	0.07	0.93
MDY	cluster_2	IJH	2026-04-10	21	4.7%	0.07	0.93

IJH	cluster_2	IJH	2026-04-10	21	4.7%	0.08	0.92
VB	cluster_2	IJH	2026-04-10	21	6.1%	0.09	0.91
PSCI	cluster_2	IJH	2026-04-13	20	2.7%	0.10	0.90
IWP	cluster_2	IJH	2026-04-15	18	1.8%	0.15	0.85
JETS	cluster_2	IJH	2026-05-06	3	-2.3%	0.17	0.83
VIS	cluster_2	IJH	2026-04-13	20	2.1%	0.24	0.76
JHMM	cluster_2	IJH	2026-04-10	21	3.9%	0.24	0.76
FIDU	cluster_2	IJH	2026-04-13	20	2.1%	0.26	0.74
NUSC	cluster_2	IJH	2026-04-13	20	3.0%	0.28	0.72
XLI	cluster_2	IJH	2026-04-13	20	1.3%	0.34	0.66
FTXR	cluster_2	IJH	2026-04-17	16	-0.5%	0.41	0.59
PKB	cluster_2	IJH	2026-04-13	20	-0.2%	0.63	0.37
IYJ	cluster_2	IJH	2026-04-13	20	0.7%	0.65	0.35
IYT	cluster_2	IJH	2026-04-14	19	1.4%	0.82	0.18
FXR	cluster_2	IJH	2026-04-14	19	-0.4%	0.85	0.15
FIW	cluster_2	IJH	2026-05-08	1	-1.0%	0.99	-0.24
XHB	cluster_2	IJH	2026-05-06	3	-3.9%	0.98	-0.28
FXD	cluster_2	IJH	2026-05-06	3	-3.2%	0.99	-0.39
NURE	cluster_14	VNQ	2026-04-13	20	5.7%	0.00	1.00
REZ	cluster_14	VNQ	2026-04-14	19	3.8%	0.01	0.99
USRT	cluster_14	VNQ	2026-04-09	22	5.2%	0.11	0.89
PSR	cluster_14	VNQ	2026-04-09	22	4.9%	0.12	0.88
RWR	cluster_14	VNQ	2026-04-10	21	4.4%	0.16	0.84
SCHH	cluster_14	VNQ	2026-04-09	22	4.4%	0.19	0.81
ICF	cluster_14	VNQ	2026-04-08	23	5.6%	0.19	0.81
FREL	cluster_14	VNQ	2026-04-10	21	4.0%	0.26	0.74
XLRE	cluster_14	VNQ	2026-04-09	22	4.3%	0.27	0.73
VNQ	cluster_14	VNQ	2026-04-10	21	4.0%	0.29	0.71
REET	cluster_14	VNQ	2026-04-13	20	3.4%	0.32	0.68
IYR	cluster_14	VNQ	2026-04-10	21	3.9%	0.32	0.68
RWO	cluster_14	VNQ	2026-04-13	20	3.0%	0.34	0.66
INDS	cluster_14	VNQ	2026-04-14	19	2.0%	0.51	0.49
SCHD	cluster_10	SCHD	2026-04-21	14	2.5%	0.00	1.00
RDIV	cluster_10	SCHD	2026-04-21	14	-0.2%	0.01	0.99
SPYD	cluster_10	SCHD	2026-04-21	14	1.0%	0.01	0.99
AML	cluster_10	SCHD	2026-04-29	8	0.6%	0.01	0.94
RPV	cluster_10	SCHD	2026-04-20	15	0.3%	0.08	0.92
DIV	cluster_10	SCHD	2026-04-07	24	1.3%	0.12	0.88
MLPA	cluster_10	SCHD	2026-04-29	8	-1.5%	0.03	0.83
HDV	cluster_10	SCHD	2026-05-01	6	-1.0%	0.14	0.73
SDOG	cluster_10	SCHD	2026-04-07	24	1.7%	0.28	0.72
DHS	cluster_10	SCHD	2026-04-20	15	-0.1%	0.30	0.70
PEY	cluster_10	SCHD	2026-04-13	20	2.3%	0.33	0.67
SPHD	cluster_10	SCHD	2026-05-05	4	-0.5%	0.30	0.64
MLPX	cluster_10	SCHD	2026-04-30	7	-3.5%	0.07	0.57

ONEY	cluster_10	SCHD	2026-04-10	21	2.1%	0.52	0.48
DVY	cluster_10	SCHD	2026-04-08	23	-0.6%	0.82	0.18
CDC	cluster_10	SCHD	2026-04-09	22	-0.6%	0.87	0.12
EMLP	cluster_10	SCHD	2026-04-28	9	-0.6%	0.56	-0.30
TDDT	cluster_19	VTIP	2026-04-01	27	1.2%	0.12	0.88
STPZ	cluster_19	VTIP	2026-04-02	26	0.9%	0.14	0.86
TDTF	cluster_19	VTIP	2026-04-09	22	1.0%	0.19	0.81
TIPX	cluster_19	VTIP	2026-04-07	24	0.8%	0.21	0.79
TIP	cluster_19	VTIP	2026-04-10	21	0.8%	0.22	0.78
SCHP	cluster_19	VTIP	2026-04-13	20	0.6%	0.23	0.77
SPIP	cluster_19	VTIP	2026-04-10	21	0.7%	0.27	0.73
LTPZ	cluster_19	VTIP	2026-05-11	0	0.0%	0.58	0.29
ENZL	cluster_1	VEA	2026-05-06	3	0.4%	0.01	0.99
VIDI	cluster_1	VEA	2026-04-09	22	6.1%	0.02	0.98
FNDF	cluster_1	VEA	2026-04-08	23	4.2%	0.02	0.98
IQDF	cluster_1	VEA	2026-04-10	21	5.0%	0.03	0.97
PXF	cluster_1	VEA	2026-04-08	23	4.2%	0.03	0.97
GWX	cluster_1	VEA	2026-04-13	20	5.1%	0.03	0.97
VEU	cluster_1	VEA	2026-04-10	21	5.0%	0.04	0.96
HFXI	cluster_1	VEA	2026-04-09	22	4.7%	0.05	0.95
VXUS	cluster_1	VEA	2026-04-10	21	5.1%	0.05	0.95
IXUS	cluster_1	VEA	2026-04-10	21	5.1%	0.06	0.94
CWI	cluster_1	VEA	2026-04-10	21	4.6%	0.06	0.94
SPDW	cluster_1	VEA	2026-04-10	21	4.6%	0.07	0.93
SCHF	cluster_1	VEA	2026-04-10	21	4.6%	0.07	0.93
ACWX	cluster_1	VEA	2026-04-10	21	4.7%	0.07	0.93
VEA	cluster_1	VEA	2026-04-10	21	4.5%	0.08	0.92
VSS	cluster_1	VEA	2026-04-13	20	4.8%	0.08	0.92
FXA	cluster_1	VEA	2026-04-13	20	2.1%	0.10	0.90
EWN	cluster_1	VEA	2026-04-10	21	4.9%	0.11	0.89
IPKW	cluster_1	VEA	2026-04-09	22	2.5%	0.13	0.87
FNDC	cluster_1	VEA	2026-04-10	21	3.7%	0.13	0.87
SCZ	cluster_1	VEA	2026-04-10	21	3.7%	0.16	0.84
SCHC	cluster_1	VEA	2026-04-13	20	3.4%	0.20	0.80
DLS	cluster_1	VEA	2026-04-13	20	2.2%	0.20	0.80
IDV	cluster_1	VEA	2026-04-06	25	4.4%	0.26	0.74
FXC	cluster_1	VEA	2026-04-20	15	-0.2%	0.27	0.73
FEP	cluster_1	VEA	2026-04-09	22	2.7%	0.28	0.72
EWI	cluster_1	VEA	2026-04-08	23	4.7%	0.28	0.72
EWG	cluster_1	VEA	2026-05-05	4	3.4%	0.33	0.67
VNQI	cluster_1	VEA	2026-05-04	5	2.6%	0.41	0.59
IMTM	cluster_1	VEA	2026-04-10	21	2.7%	0.41	0.59
EWG	cluster_1	VEA	2026-04-14	19	0.8%	0.41	0.59
EWO	cluster_1	VEA	2026-04-10	21	4.2%	0.43	0.57
VIGI	cluster_1	VEA	2026-04-14	19	0.5%	0.43	0.57

NFRA	cluster_1	VEA	2026-04-09	22	-0.0%	0.46	0.54
VYMI	cluster_1	VEA	2026-04-09	22	1.7%	0.46	0.54
BBAX	cluster_1	VEA	2026-04-30	7	0.9%	0.51	0.49
DWX	cluster_1	VEA	2026-05-04	5	1.0%	0.54	0.46
EWL	cluster_1	VEA	2026-04-15	18	-0.2%	0.57	0.43
PID	cluster_1	VEA	2026-04-30	7	-0.0%	0.58	0.41
IHDG	cluster_1	VEA	2026-05-06	3	-1.5%	0.69	0.30
RWX	cluster_1	VEA	2026-05-06	3	-1.4%	0.77	0.22
IDLV	cluster_1	VEA	2026-04-30	7	-0.6%	0.78	0.21
EWC	cluster_1	VEA	2026-04-09	22	3.0%	0.81	0.19
EZA	cluster_1	VEA	2026-05-08	1	0.1%	0.67	0.18
EBND	cluster_1	VEA	2026-05-06	3	0.0%	0.81	0.18
EMLC	cluster_1	VEA	2026-05-06	3	-0.0%	0.84	0.15
HDEF	cluster_1	VEA	2026-04-07	24	0.5%	0.93	0.01
EWD	cluster_1	VEA	2026-05-06	3	-2.6%	0.96	-0.15
EWU	cluster_1	VEA	2026-04-30	7	-1.1%	0.98	-0.35
IGF	cluster_1	VEA	2026-04-30	7	-1.6%	0.76	-0.35
SLX	cluster_12	URA + GUNR	2026-04-09	22	10.8%	0.00	1.00
XME	cluster_12	URA + GUNR	2026-04-15	18	7.7%	0.04	0.96
DBB	cluster_12	URA + GUNR	2026-04-10	21	6.8%	0.25	0.75
URA	cluster_12	URA + GUNR	2026-04-14	19	8.3%	0.35	0.65
COPX	cluster_12	URA + GUNR	2026-05-08	1	1.9%	0.55	0.42
NORW	cluster_12	URA + GUNR	2025-12-09	104	31.8%	0.63	0.32
NANR	cluster_12	URA + GUNR	2026-05-11	0	0.0%	0.70	-0.35
IGE	cluster_12	URA + GUNR	2026-05-01	6	0.1%	0.64	-0.59
MOO	cluster_12	URA + GUNR	2026-05-06	3	-2.8%	0.86	-0.82
PFXF	cluster_9	HYG	2026-04-14	19	4.1%	0.00	1.00
PFF	cluster_9	HYG	2026-04-15	18	1.5%	0.05	0.95
HYEM	cluster_9	HYG	2026-04-13	20	1.3%	0.12	0.88
PSK	cluster_9	HYG	2026-04-17	16	0.2%	0.41	0.59
VRP	cluster_9	HYG	2026-04-10	21	0.5%	0.45	0.55
FPE	cluster_9	HYG	2026-04-14	19	0.6%	0.46	0.54
PGX	cluster_9	HYG	2026-04-17	16	-0.4%	0.58	0.42
HYLS	cluster_9	HYG	2026-04-09	22	0.5%	0.64	0.36
SHYG	cluster_9	HYG	2026-04-07	24	0.9%	0.76	0.24
FALN	cluster_9	HYG	2026-04-10	21	0.6%	0.84	0.16
JNK	cluster_9	HYG	2026-04-08	23	0.4%	0.87	0.13
SJNK	cluster_9	HYG	2026-04-07	24	0.9%	0.89	0.11
HYS	cluster_9	HYG	2026-04-08	23	0.1%	0.89	0.11
BBHY	cluster_9	HYG	2026-04-08	23	0.3%	0.90	0.10
HYG	cluster_9	HYG	2026-04-08	23	0.3%	0.91	0.09
HYLB	cluster_9	HYG	2026-04-08	23	0.2%	0.92	0.08
ANGL	cluster_9	HYG	2026-04-09	22	0.2%	0.95	0.05
PHB	cluster_9	HYG	2026-04-10	21			
IWN	cluster_7	IJR	2026-04-09	22	5.8%	0.01	0.99

VTWV	cluster_7	IJR	2026-04-09	22	5.7%	0.01	0.99
OMFS	cluster_7	IJR	2026-04-09	22	6.3%	0.02	0.98
ISMD	cluster_7	IJR	2026-04-09	22	7.5%	0.02	0.98
EES	cluster_7	IJR	2026-04-08	23	5.1%	0.02	0.98
VIOO	cluster_7	IJR	2026-04-09	22	5.0%	0.04	0.96
IJR	cluster_7	IJR	2026-04-09	22	5.0%	0.05	0.95
SPSM	cluster_7	IJR	2026-04-09	22	5.0%	0.05	0.95
PRFZ	cluster_7	IJR	2026-04-10	21	6.8%	0.05	0.95
FNDA	cluster_7	IJR	2026-04-10	21	5.4%	0.09	0.91
FYX	cluster_7	IJR	2026-04-08	23	6.5%	0.11	0.89
FNX	cluster_7	IJR	2026-04-10	21	3.9%	0.12	0.88
RWJ	cluster_7	IJR	2026-04-09	22	4.6%	0.15	0.85
SLYV	cluster_7	IJR	2026-04-10	21	4.5%	0.16	0.84
IJS	cluster_7	IJR	2026-04-10	21	4.5%	0.17	0.83
RFV	cluster_7	IJR	2026-04-10	21	2.9%	0.17	0.83
VIOV	cluster_7	IJR	2026-04-10	21	4.5%	0.18	0.82
EZM	cluster_7	IJR	2026-04-10	21	3.7%	0.23	0.77
VBR	cluster_7	IJR	2026-04-10	21	3.3%	0.32	0.68
DES	cluster_7	IJR	2026-04-08	23	3.1%	0.39	0.61
PSP	cluster_7	IJR	2026-04-15	18	-1.1%	0.42	0.58
KCE	cluster_7	IJR	2026-04-13	20	4.7%	0.49	0.51
RWK	cluster_7	IJR	2026-04-09	22	2.7%	0.51	0.49
RZV	cluster_7	IJR	2026-04-13	20	3.5%	0.58	0.42
IJJ	cluster_7	IJR	2026-04-10	21	2.2%	0.65	0.35
IVOV	cluster_7	IJR	2026-04-10	21	2.1%	0.65	0.35
MDYV	cluster_7	IJR	2026-04-10	21	2.2%	0.65	0.35
DON	cluster_7	IJR	2026-04-13	20	0.9%	0.72	0.28
IAI	cluster_7	IJR	2026-04-10	21	4.6%	0.76	0.24
FYT	cluster_7	IJR	2026-04-07	24	4.3%	0.82	0.18
XSLV	cluster_7	IJR	2026-04-09	22	-0.6%	0.93	0.07
SDVY	cluster_7	IJR	2026-04-10	21	1.0%	0.93	0.07
KRE	cluster_7	IJR	2026-04-09	22	-1.9%	0.93	0.07
SMDV	cluster_7	IJR	2026-04-09	22	-1.0%	0.94	0.06
REM	cluster_7	IJR	2026-04-13	20	0.1%	0.95	0.05
KBE	cluster_7	IJR	2026-04-09	22	-1.1%	0.97	0.03
FNCL	cluster_7	IJR	2026-04-13	20	-0.8%	0.99	0.00
XLF	cluster_7	IJR	2026-04-13	20	-0.9%	1.00	0.00
REGL	cluster_7	IJR	2026-04-10	21	-0.9%	1.00	0.00
IAT	cluster_7	IJR	2026-04-10	21	-1.4%	1.00	0.00
SMMV	cluster_7	IJR	2026-04-09	22	-1.2%	1.00	-0.06
HYD	cluster_16	MUB	2026-04-08	23	0.9%	0.50	0.50
HYMB	cluster_16	MUB	2026-04-09	22	0.5%	0.53	0.47
FMB	cluster_16	MUB	2026-04-17	16	0.0%	0.67	0.33
NYF	cluster_16	MUB	2026-04-15	18	0.4%	0.69	0.31
PZA	cluster_16	MUB	2026-04-10	21	0.6%	0.73	0.27

ITM	cluster_16	MUB	2026-05-08	1	0.0%	0.75	0.24
MUB	cluster_16	MUB	2026-04-17	16	-0.1%	0.77	0.23
CMF	cluster_16	MUB	2026-05-06	3	0.0%	0.86	0.14
TFI	cluster_16	MUB	2026-05-06	3	-0.0%	0.89	0.11
SUB	cluster_16	MUB	2026-05-11	0	0.0%	0.90	0.00
EPI	cluster_25	INDA + INDY + EPI + PIN	2026-04-17	16	-5.1%	0.83	0.17
EWM	cluster_18	EWZ + ILF + DBA + ECH	2026-04-22	13	3.3%	0.00	1.00
VNM	cluster_18	EWZ + ILF + DBA + ECH	2026-04-13	20	6.5%	0.07	0.93
TUR	cluster_18	EWZ + ILF + DBA + ECH	2026-04-10	21	3.9%	0.31	0.69
ECH	cluster_18	EWZ + ILF + DBA + ECH	2026-05-06	3	-2.0%	0.98	-0.17
RSX	cluster_18	EWZ + ILF + DBA + ECH	2023-07-27	699			
IHF	cluster_11	VHT	2026-05-08	1	0.2%	0.00	1.00
PSCH	cluster_11	VHT	2026-04-13	20	4.8%	0.06	0.94
ARKG	cluster_11	VHT	2026-04-15	18	0.0%	0.45	0.55
FXH	cluster_11	VHT	2026-04-10	21	0.5%	0.78	0.22
IBB	cluster_11	VHT	2026-05-06	3	-2.2%	0.81	-0.01
LDUR	cluster_5	BND	2026-04-07	24	0.4%	0.29	0.71
CMBS	cluster_5	BND	2026-05-08	1	-0.1%	0.41	0.57
IAGG	cluster_5	BND	2026-05-08	1	-0.3%	0.42	0.53
BNDX	cluster_5	BND	2026-05-08	1	-0.3%	0.41	0.53
SPSB	cluster_5	BND	2026-04-09	22	0.1%	0.71	0.29
IGLB	cluster_5	BND	2026-05-08	1	-0.4%	0.76	0.20
USIG	cluster_5	BND	2026-05-06	3	-0.2%	0.81	0.17
VCLT	cluster_5	BND	2026-05-08	1	-0.4%	0.78	0.17
GTO	cluster_5	BND	2026-05-07	2	0.0%	0.80	0.17
FLCO	cluster_5	BND	2026-05-07	2	0.3%	0.81	0.16
SPLB	cluster_5	BND	2026-05-08	1	-0.3%	0.78	0.16
CORP	cluster_5	BND	2026-05-06	3	-0.2%	0.83	0.15
GNMA	cluster_5	BND	2026-05-07	2	-0.1%	0.79	0.11
BAB	cluster_5	BND	2026-05-11	0	0.0%	0.65	0.10
IGIB	cluster_5	BND	2026-05-07	2	0.1%	0.87	0.09
SPIB	cluster_5	BND	2026-05-06	3	-0.1%	0.90	0.08
LQD	cluster_5	BND	2026-05-08	1	-0.2%	0.85	0.07
AGGY	cluster_5	BND	2026-05-08	1	-0.3%	0.84	0.04
VCIT	cluster_5	BND	2026-05-07	2	0.1%	0.90	0.03
IUSB	cluster_5	BND	2026-05-08	1	-0.2%	0.87	0.02
BSV	cluster_5	BND	2026-05-06	3	-0.1%	0.91	0.02
QLTA	cluster_5	BND	2026-05-08	1	-0.2%	0.90	0.00
FBND	cluster_5	BND	2026-05-08	1	-0.3%	0.87	-0.05
VMBS	cluster_5	BND	2026-05-11	0	0.0%	0.86	-0.08
BOND	cluster_5	BND	2026-05-11	0	0.0%	0.91	-0.10
MBB	cluster_5	BND	2026-05-11	0	0.0%	0.87	-0.17
GVI	cluster_5	BND	2026-05-11	0	0.0%	0.93	-0.22
AGG	cluster_5	BND	2026-05-11	0	0.0%	0.89	-0.23
BND	cluster_5	BND	2026-05-11	0	0.0%	0.90	-0.26

SPAB	cluster_5	BND	2026-05-11	0	0.0%	0.89	-0.30
SCHZ	cluster_5	BND	2026-05-11	0	0.0%	0.93	-0.34
SPMB	cluster_5	BND	2026-05-11	0	0.0%	0.89	-0.36
AGZ	cluster_5	BND	2026-05-11	0	0.0%	0.98	-0.53
SIVR	cluster_15	GLD + IAU + SLV	2026-05-11	0	0.0%	0.15	0.80
SLV	cluster_15	GLD + IAU + SLV	2026-05-11	0	0.0%	0.15	0.79
PPLT	cluster_15	GLD + IAU + SLV	2026-05-11	0	0.0%	0.50	0.16
SILJ	cluster_15	GLD + IAU + SLV	2026-05-11	0	0.0%	0.52	0.14
SIL	cluster_15	GLD + IAU + SLV	2026-05-11	0	0.0%	0.59	0.10
XLP	cluster_17	XLU + XLP	2026-05-05	4	-0.8%	0.00	1.00
KXI	cluster_17	XLU + XLP	2026-05-06	3	-1.6%	0.02	0.98
IYK	cluster_17	XLU + XLP	2026-05-06	3	-0.2%	0.01	0.97
VDC	cluster_17	XLU + XLP	2026-05-01	6	-1.7%	0.04	0.96
FSTA	cluster_17	XLU + XLP	2026-05-01	6	-1.8%	0.04	0.96
UUP	cluster_26	SQQQ + UUP	2026-04-27	10	-0.4%	0.47	-0.47

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
NXTG	cluster_3	QQQ	1	1.00
VLUE	cluster_6	VTV	1	1.00
IBDR	cluster_5	BND	1	1.00
MINT	cluster_23	SHV + MINT + GBIL + ICSH	1	1.00
FLTR	cluster_22	FLOT + BKLN + SRLN + FLRN...	1	1.00
IHF	cluster_11	VHT	1	1.00
EWT	cluster_8	IEMG	1	1.00
EWM	cluster_18	EWZ + ILF + DBA + ECH	1	1.00
FV	cluster_2	IJH	1	1.00
EIS	cluster_4	VOO	1	1.00
CQQQ	cluster_13	KWEB	1	1.00
VPL	cluster_20	EWJ	1	1.00
SLX	cluster_12	URA + GUNR	1	1.00
FXV	cluster_24	BWX	1	1.00
FTGC	cluster_21	PDBC	1	1.00
XLP	cluster_17	XLU + XLP	1	1.00
NURE	cluster_14	VNQ	1	1.00
SCHD	cluster_10	SCHD	1	1.00
PFXF	cluster_9	HYG	1	1.00
ENZL	cluster_1	VEA	1	0.99

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
IHI	cluster_11	VHT	1.00	-1.00
SQQQ	cluster_26	SQQQ + UUP	1.00	-1.00
SPLV	cluster_6	VTV	1.00	-1.00
EIDO	cluster_18	EWZ + ILF + DBA + ECH	1.00	-1.00
IDU	cluster_17	XLU + XLP	0.99	-0.99
LVHD	cluster_10	SCHD	0.99	-0.95

PPA	cluster_2	IJH	0.93	-0.93
IEF	cluster_5	BND	0.96	-0.93
MOO	cluster_12	URA + GUNR	0.86	-0.82
GOAU	cluster_15	GLD + IAU + SLV	0.88	-0.79
EFAV	cluster_1	VEA	0.94	-0.72
SHM	cluster_16	MUB	0.96	-0.70
ESPO	cluster_4	VOO	0.97	-0.56
KIE	cluster_7	IJR	1.00	-0.44
CHIQ	cluster_13	KWEB	0.90	-0.33
INDY	cluster_25	INDA + INDY + EPI + PIN	0.98	-0.31
IBUY	cluster_3	QQQ	0.98	-0.11
ANGL	cluster_9	HYG	0.95	0.05
LTPZ	cluster_19	VTIP	0.58	0.29
FXE	cluster_24	BWX	0.65	0.35

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
RSX			cluster_18	11	LONG	0.371	0.052	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_18	11	FLAT	0.492	0.081	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.692	0.040	0.00007	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.635	0.221	0.00034	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_24	6	FLAT	0.732	0.094	0.00022	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_18	11	FLAT	0.591	0.174	0.00038	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.787	0.130	0.00030	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.825	0.021	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.825	0.089	0.00022	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.720	0.182	0.00041	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.656	0.188	0.00047	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.793	0.162	0.00038	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	11	LONG	0.487	0.219	0.00056	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.793	0.179	0.00043	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	LONG	0.844	0.124	0.00033	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_24	6	LONG	0.818	0.203	0.00047	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	LONG	0.876	0.160	0.00030	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.836	0.231	0.00044	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.250	0.00044	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_18	11	LONG	0.452	0.334	0.00082	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_15	13	LONG	0.750	0.269	0.00072	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.877	0.263	0.00051	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.804	0.271	0.00070	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	LONG	0.751	0.357	0.00082	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	11	FLAT	0.514	0.374	0.00105	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_18	11	FLAT	0.578	0.409	0.00105	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	LONG	0.855	0.254	0.00070	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_22	9	FLAT	0.864	0.332	0.00070	
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_18	11	LONG	0.637	0.429	0.00116	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	LONG	0.905	0.242	0.00062	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20